US012414359B2

(12) **United States Patent**
Leng

(10) **Patent No.: US 12,414,359 B2**
(45) **Date of Patent: *Sep. 9, 2025**

(54) **INTEGRATED CIRCUIT STRUCTURE INCLUDING A METAL-INSULATOR-METAL (MIM) CAPACITOR MODULE AND A THIN-FILM RESISTOR (TFR) MODULE**

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(72) Inventor: **Yaojian Leng**, Vancouver, WA (US)

(73) Assignee: **Microchip Technology Incorporated**, Chandler, AZ (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 663 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **17/536,597**

(22) Filed: **Nov. 29, 2021**

(65) **Prior Publication Data**

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Related U.S. Application Data

(60) Provisional application No. 63/244,366, filed on Sep. 15, 2021.

(51) **Int. Cl.**

H10D 84/80 (2025.01)
H01L 23/522 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC ***H10D 84/811*** (2025.01); ***H01L 23/5223*** (2013.01); ***H01L 23/5226*** (2013.01);

(Continued)

(58) **Field of Classification Search**

CPC H01L 23/481; H01L 21/76898; H01L 2225/06544; H01L 25/0657; H01L 28/40;

(Continued)

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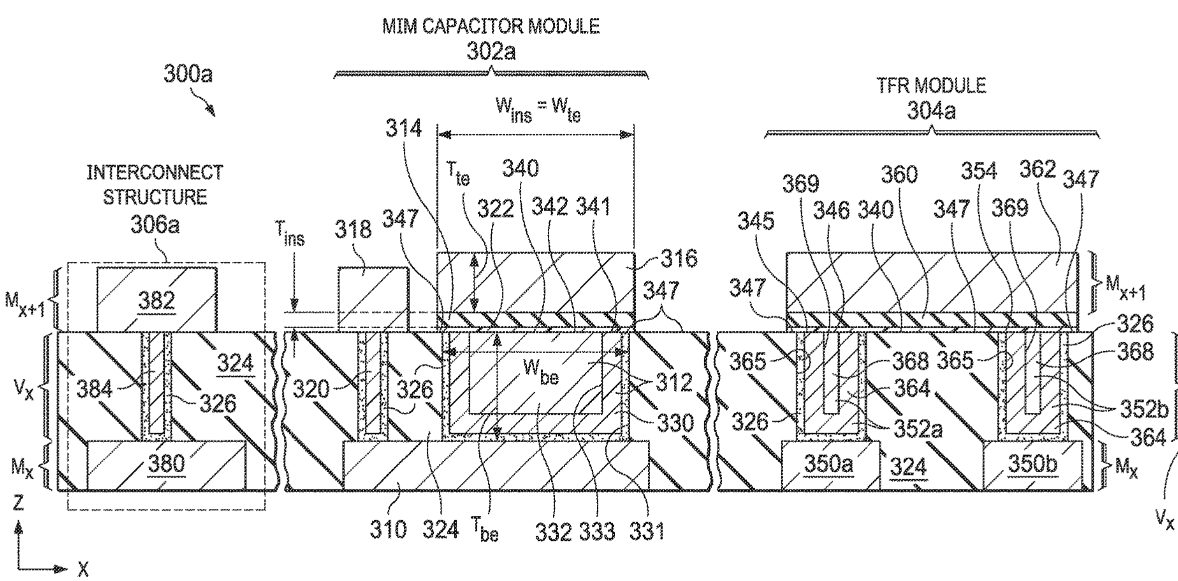
Primary Examiner — Eliseo Ramos Feliciano
Assistant Examiner — Brad A Knudson
(74) *Attorney, Agent, or Firm* — SLAYDEN GRUBERT BEARD PLLC

(57) **ABSTRACT**

An integrated circuit structure including a metal-insulator-metal (MIM) capacitor module and a thin-film resistor (TFR) module is provided. The MIM capacitor module includes a bottom electrode base formed in a lower metal layer, a bottom electrode formed in a dielectric region between the lower metal layer and an upper metal layer, an insulator formed over the bottom electrode, and a top electrode formed in the upper metal layer over the insulator. The bottom electrode includes a cup-shaped bottom electrode component and a bottom electrode fill component formed in an interior opening defined by the cup-shaped bottom electrode component. The TFR module includes a pair of metal heads formed in the dielectric region and a resistor element connected across the pair of metal heads.

(Continued)



Each metal head includes a cup-shaped head component and a head fill component formed in an interior opening defined by the cup-shaped head component.

14 Claims, 18 Drawing Sheets

(51) **Int. Cl.**

H01L 23/532	(2006.01)
H10D 1/00	(2025.01)
H10D 1/68	(2025.01)
H10D 84/01	(2025.01)
H01L 21/768	(2006.01)
H10D 1/47	(2025.01)
H10D 84/00	(2025.01)
H10D 86/80	(2025.01)

(52) **U.S. Cl.**

CPC ***H01L 23/5228*** (2013.01); ***H01L 23/53266*** (2013.01); ***H10D 1/00*** (2025.01); ***H10D 1/696*** (2025.01); ***H10D 84/01*** (2025.01); *H01L 21/7684* (2013.01); *H01L 21/76843* (2013.01); *H01L 21/76885* (2013.01); *H10D 1/47* (2025.01); *H10D 1/68* (2025.01); *H10D 1/692* (2025.01); *H10D 84/206* (2025.01); *H10D 86/80* (2025.01)

(58) **Field of Classification Search**

CPC H01L 28/20; H01L 28/60; H01L 21/7687; H01L 23/5226; H01L 28/00; H01L 21/7684; H01L 21/76843; H01L 21/76885; H01L 23/5223; H01L 23/5228; H01L 23/53266; H01L 23/647; H01L 27/016; H10D 84/206; H10D 1/00; H10D 1/696; H10D 84/01; H10D 1/47; H10D 1/68; H10D 1/692; H10D 86/80; H10D 84/811

See application file for complete search history.

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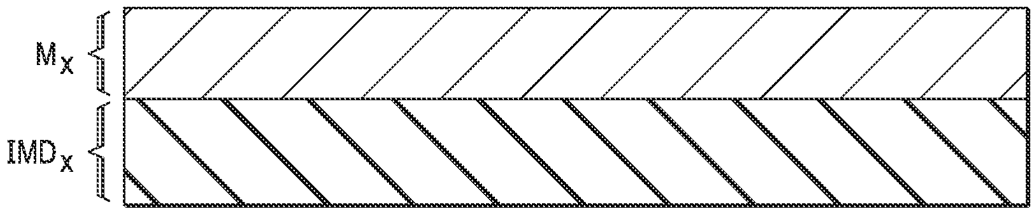


FIG. 1A
(PRIOR ART)

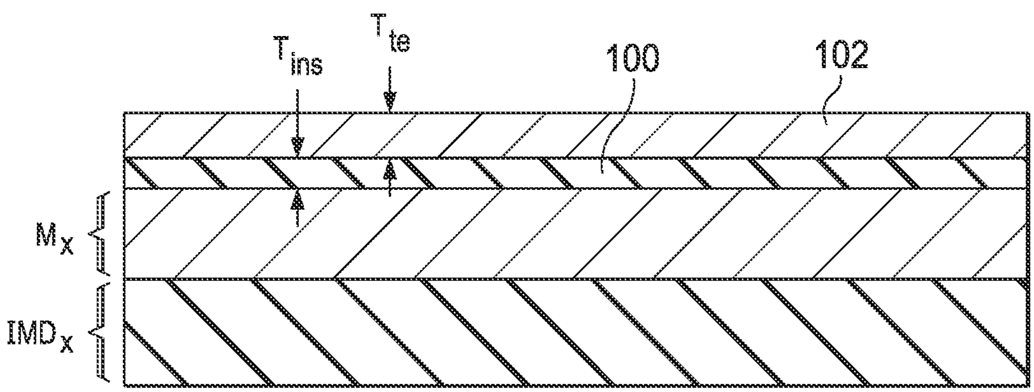


FIG. 1B
(PRIOR ART)

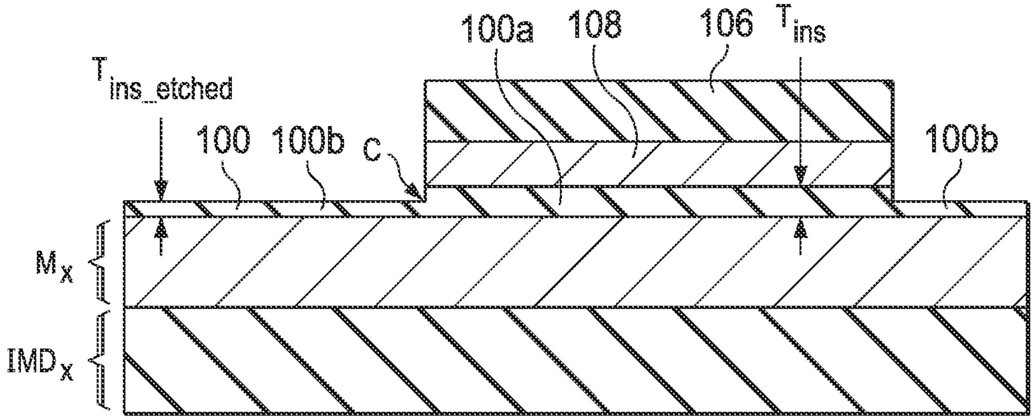


FIG. 1C
(PRIOR ART)

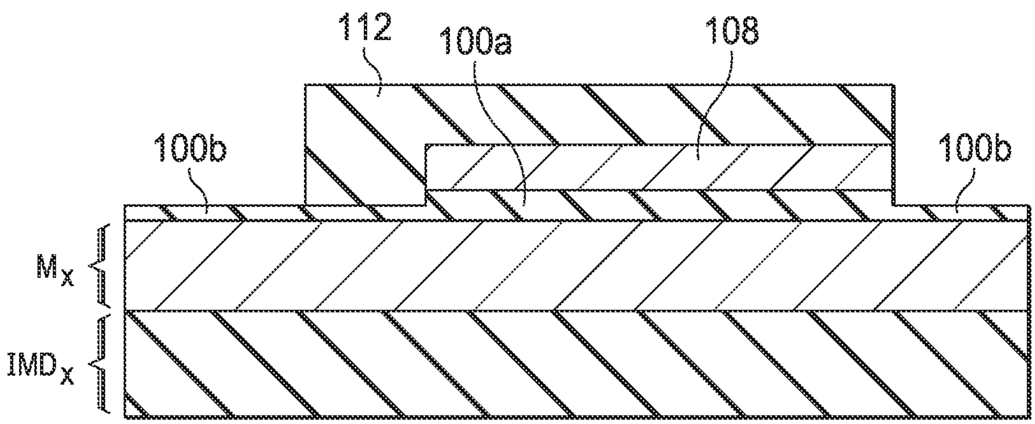


FIG. 1D
(PRIOR ART)

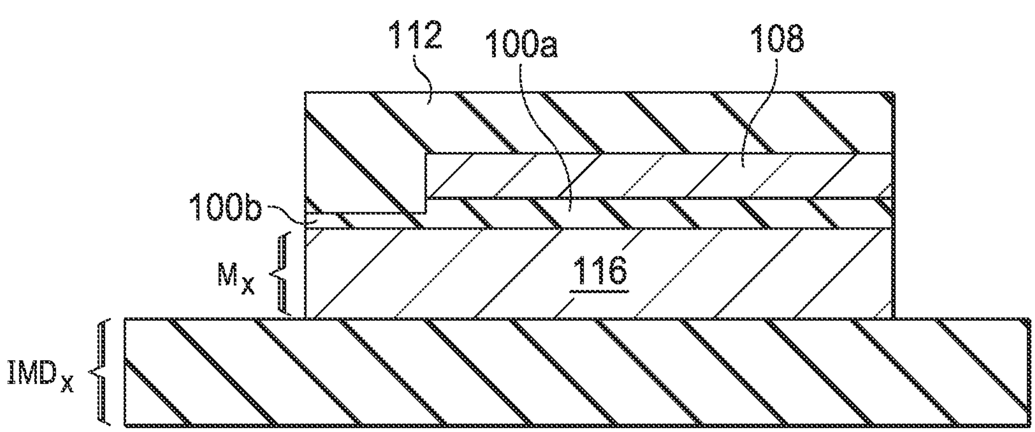


FIG. 1E
(PRIOR ART)

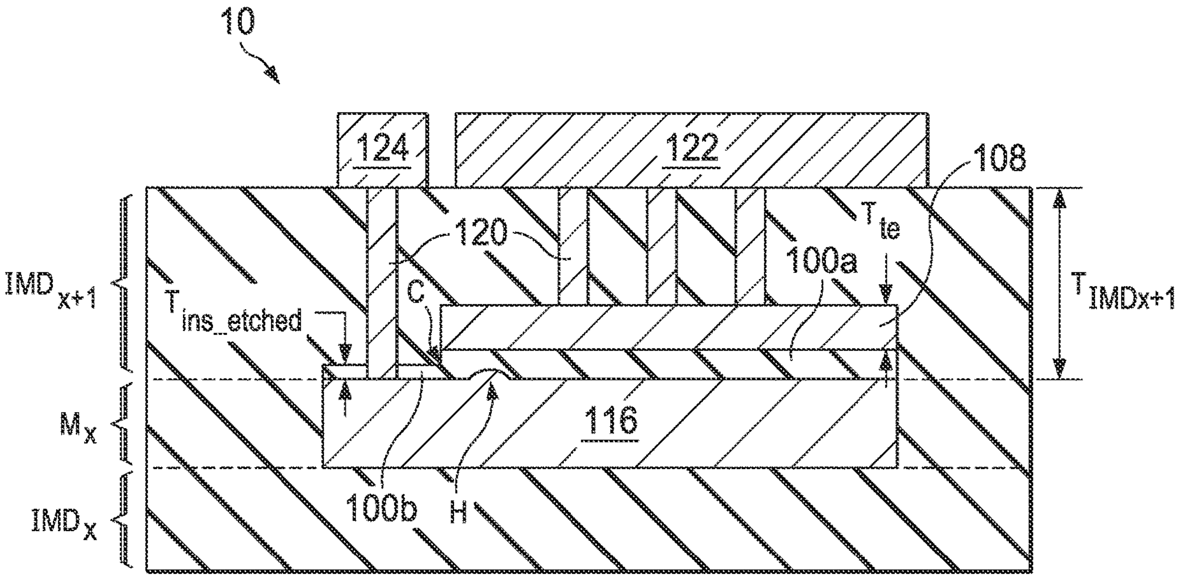


FIG. 1F
(PRIOR ART)

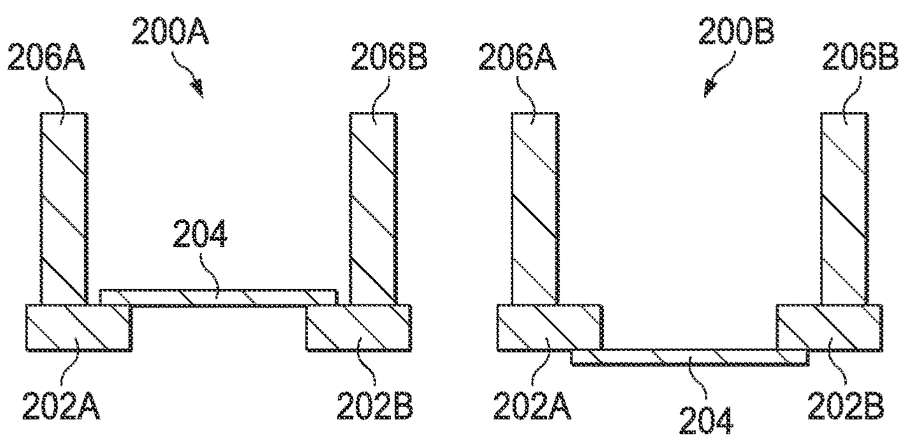
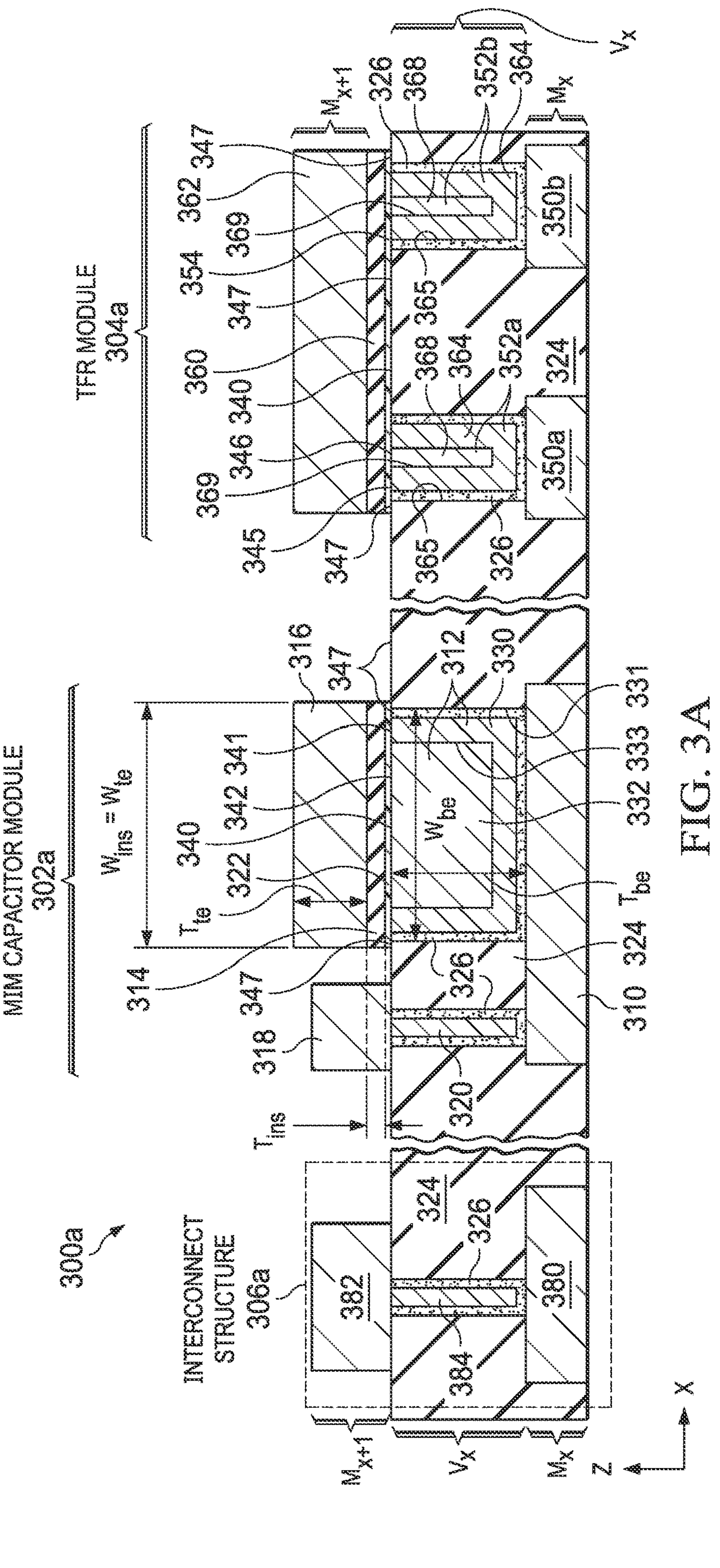


FIG. 2
(PRIOR ART)



TFR MODULE
304a
MIM CAPACITOR MODULE
302a
INTERCONNECT
STRUCTURE
306a
300a
M_{X+1}
V_X
M_X
$W_{ins} = W_{te}$
T_{te}
T_{ins}
W_{be}
T_{be}
Z
X
310
312
314
316
318
320
322
324
326
330
331
332
333
340
341
342
345
346
347
350a
350b
352a
352b
354
360
362
364
365
368
369
380
382
384

FIG. 3A

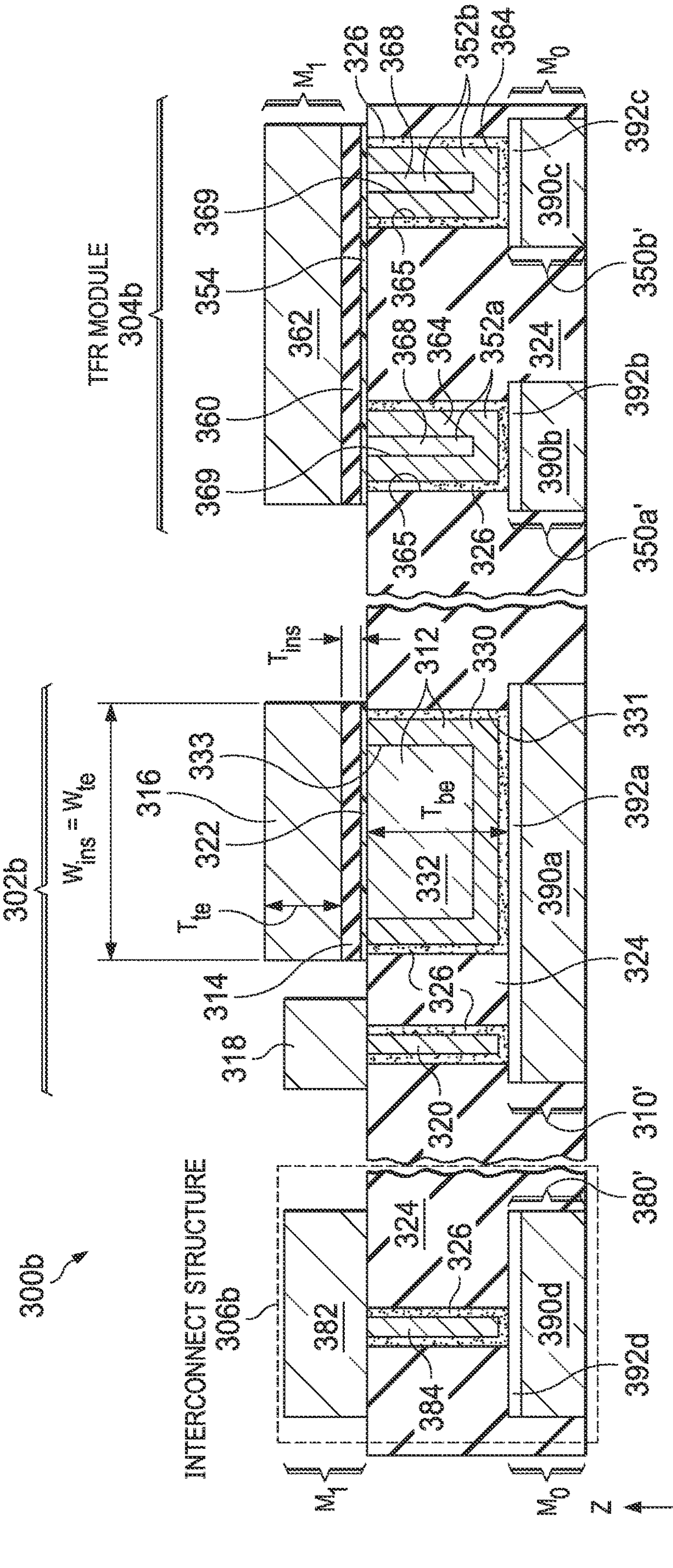
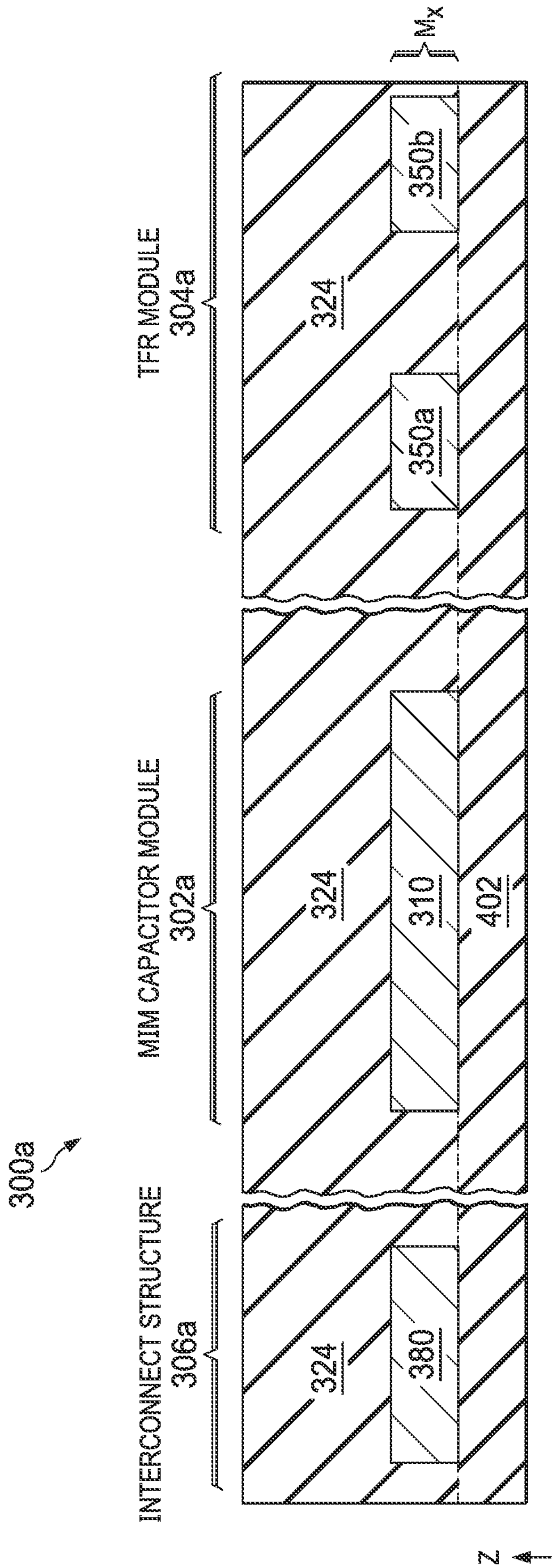

300b
INTERCONNECT STRUCTURE
306b
MIM CAPACITOR MODULE
302b
TFR MODULE
304b
M_1
M_0
382
384
324
326
390d
392d
380'
318
320
310'
314
T_{te}
$W_{ins} = W_{te}$
322
316
333
332
T_{be}
390a
392a
331
330
312
T_{ins}
369
365
360
368
364
352a
362
354
392b
390b
350a'
350b'
390c
392c
352b
Z
X

FIG. 3B



300a
INTERCONNECT STRUCTURE
306a
MIM CAPACITOR MODULE
302a
TFR MODULE
304a
M_X
324
380
310
402
350a
350b
X
Z

FIG. 4

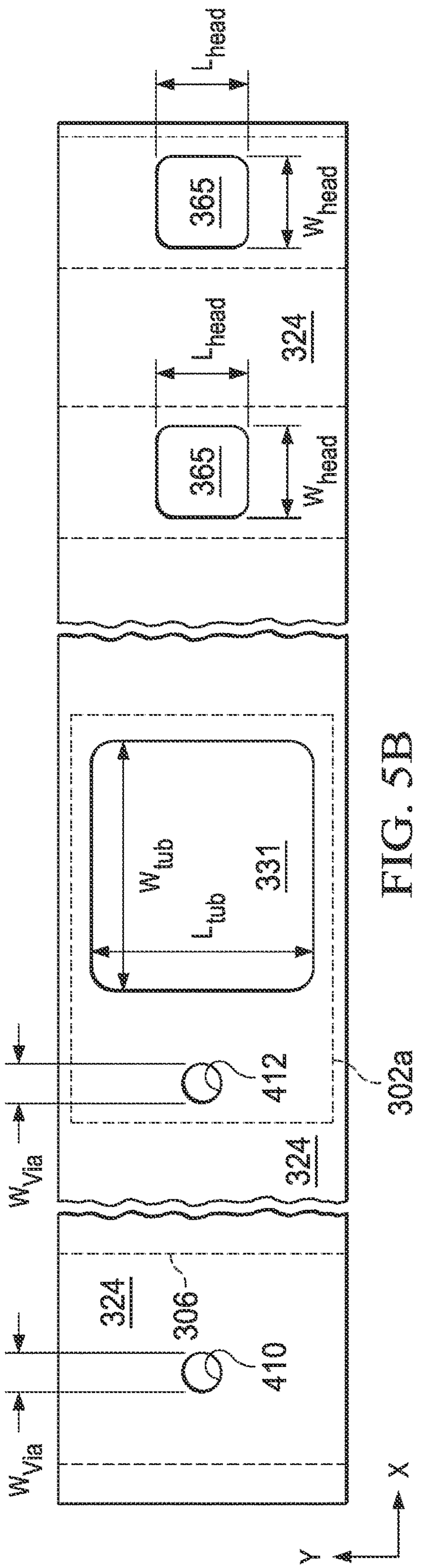

L_head
W_head
365
324
L_tub
W_tub
331
412
302a
306
410
W_Via
X
Y

FIG. 5B

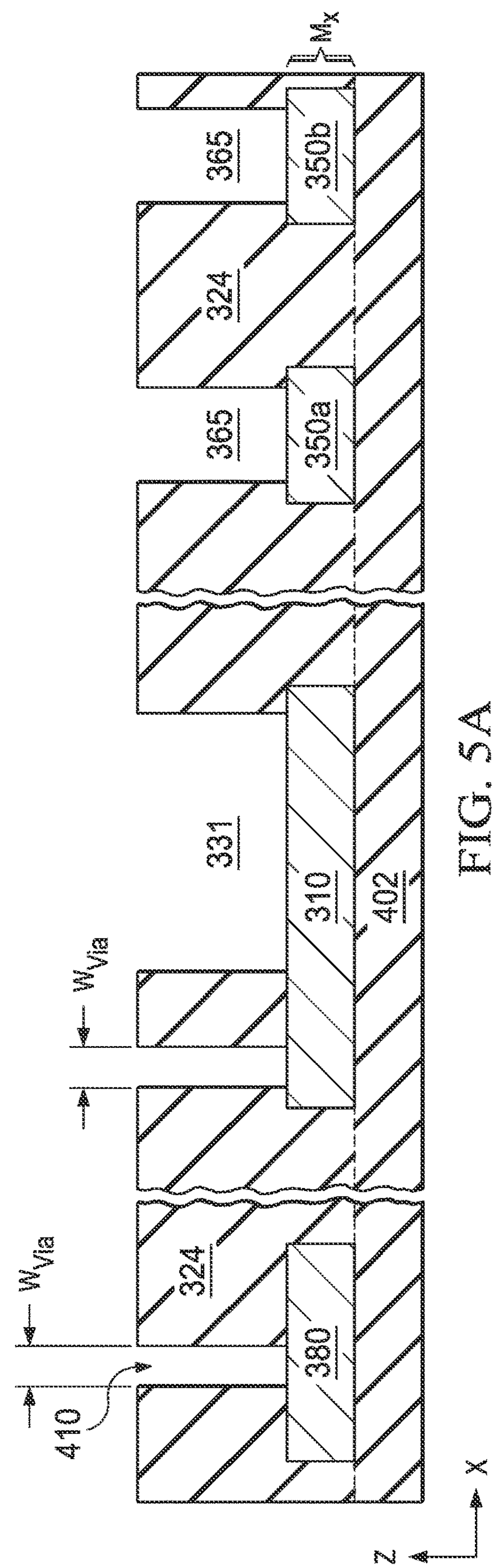

M_x
365
350b
324
350a
331
310
402
380
410
W_Via
X
Z

FIG. 5A

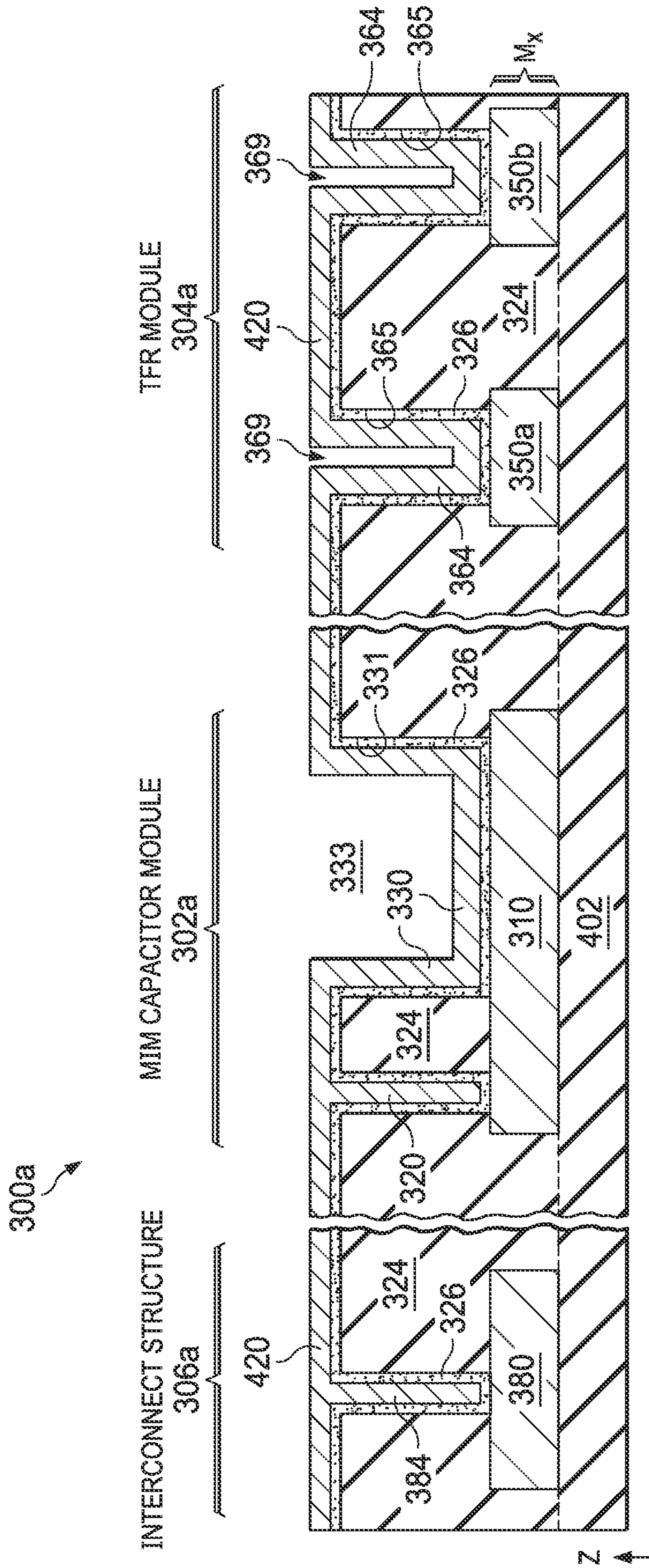

300a
TFR MODULE
304a
MIM CAPACITOR MODULE
302a
INTERCONNECT STRUCTURE
306a
364
365
Mx
369
350b
420
365
326
324
369
350a
364
331
326
333
330
310
402
324
320
420
324
326
380
384
Z
X

FIG. 6

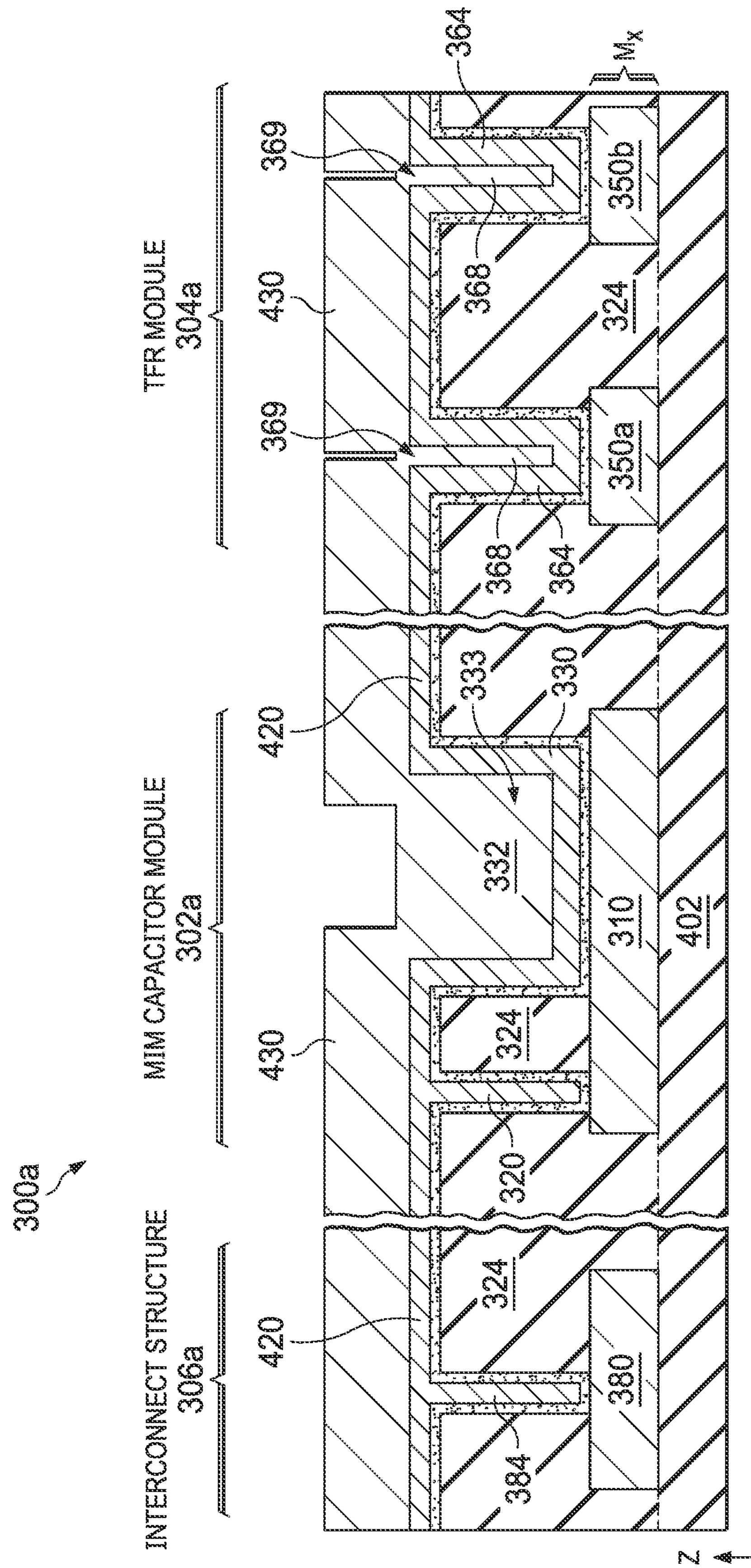

300a
INTERCONNECT STRUCTURE
306a
MIM CAPACITOR MODULE
302a
TFR MODULE
304a
420
430
369
364
368
350b
350a
324
333
330
332
310
402
320
384
380
Mx
Z
X

FIG. 7

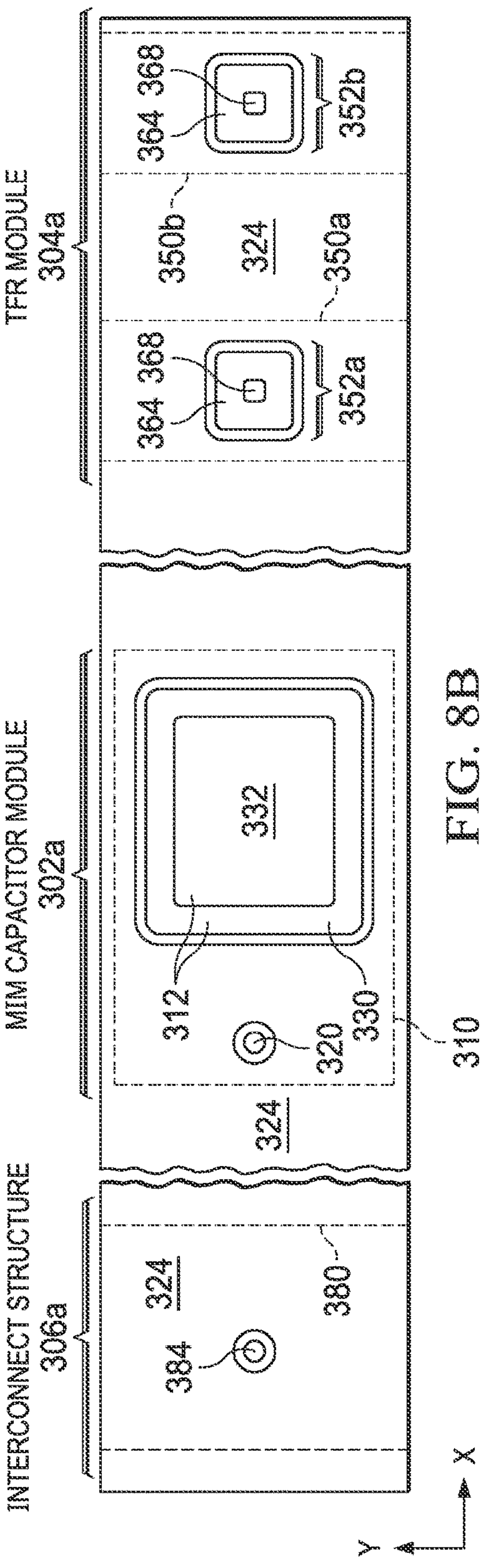

INTERCONNECT STRUCTURE
306a
MIM CAPACITOR MODULE
302a
TFR MODULE
304a
324
384
380
320
312
330
310
332
364
368
352a
350a
350b
352b
X
Y

FIG. 8B

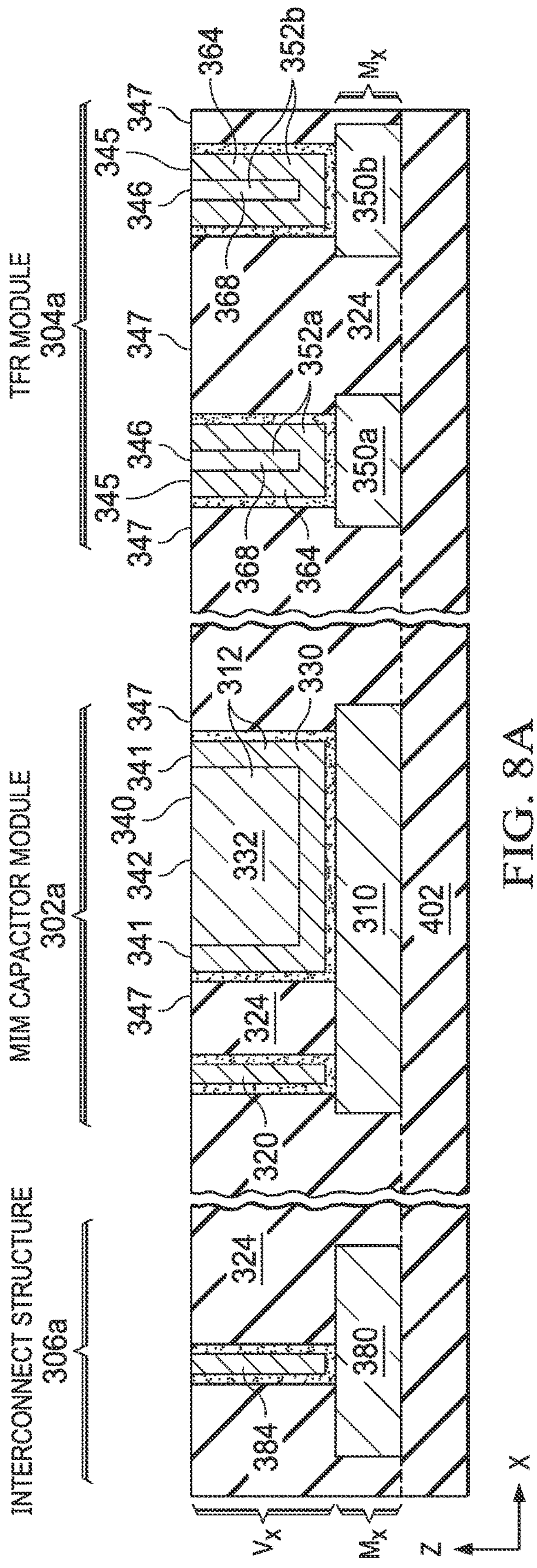

INTERCONNECT STRUCTURE
306a
MIM CAPACITOR MODULE
302a
TFR MODULE
304a
324
384
380
320
347
341
342
340
332
310
312
330
402
345
346
368
364
352a
350a
350b
352b
M_X
V_X
X
Z

FIG. 8A

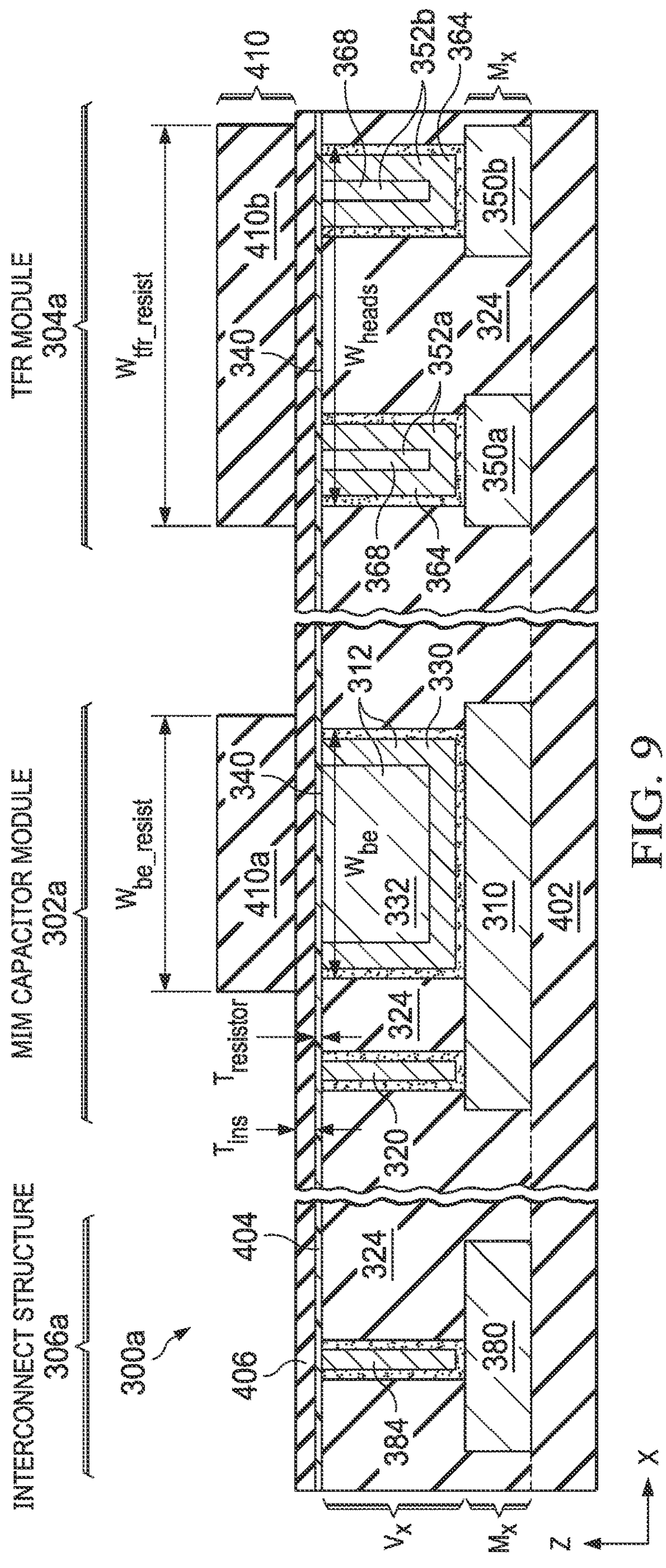

TFR MODULE
304a
MIM CAPACITOR MODULE
302a
INTERCONNECT STRUCTURE
306a
300a
410
368
352b
364
M_x
410b
350b
340
W_{tfr_resist}
W_{heads}
352a
324
350a
368
364
312
330
340
W_{be_resist}
W_{be}
410a
332
310
402
$T_{resistor}$
324
T_{ins}
320
404
324
406
380
384
V_x
M_x
Z
X

FIG. 9

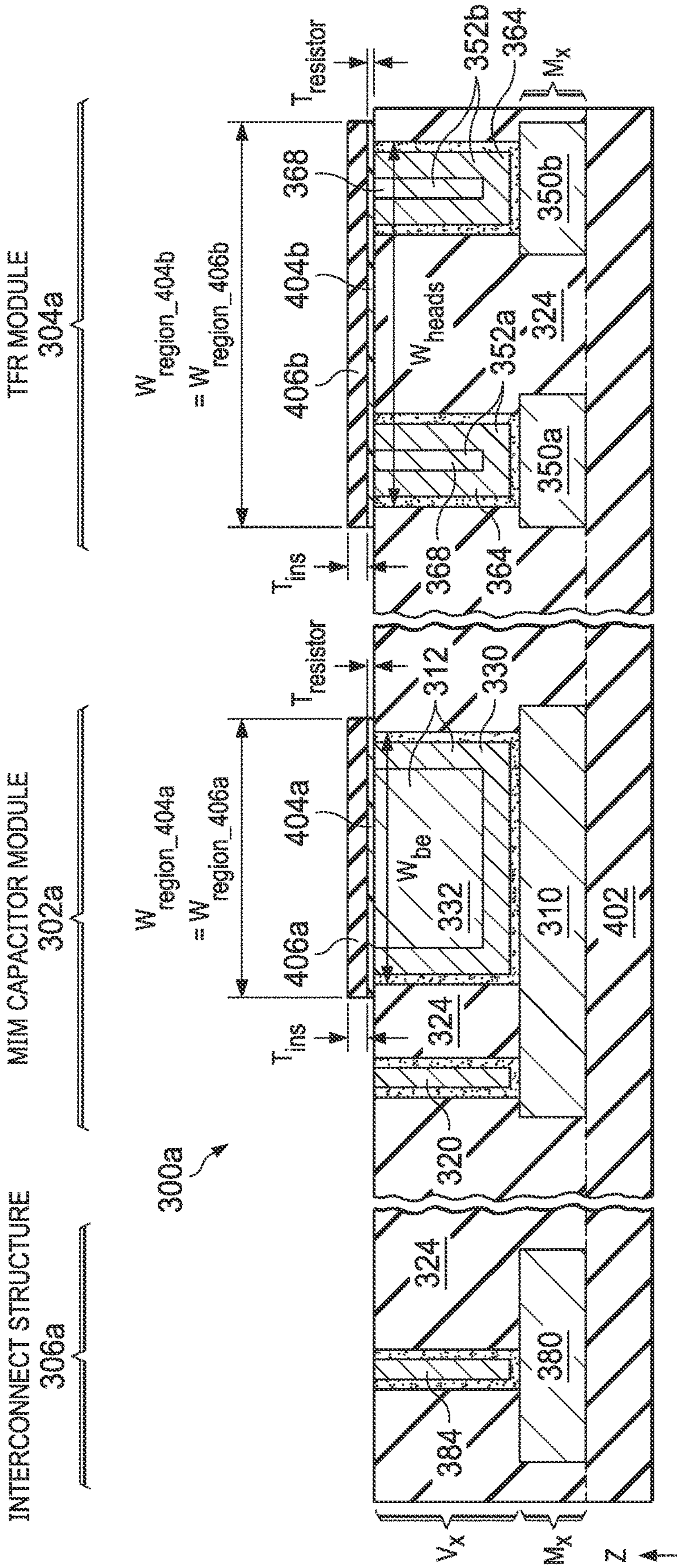

TFR MODULE
304a
MIM CAPACITOR MODULE
302a
INTERCONNECT STRUCTURE
306a
300a
$W_{region\_404b} = W_{region\_406b}$
$W_{region\_404a} = W_{region\_406a}$
$T_{resistor}$
T_{ins}
W_{heads}
W_{be}
352b
364
368
404b
406b
350b
324
352a
350a
368
364
312
330
404a
406a
332
310
402
324
320
324
380
384
V_X
M_X
M_X
Z
X

FIG. 10

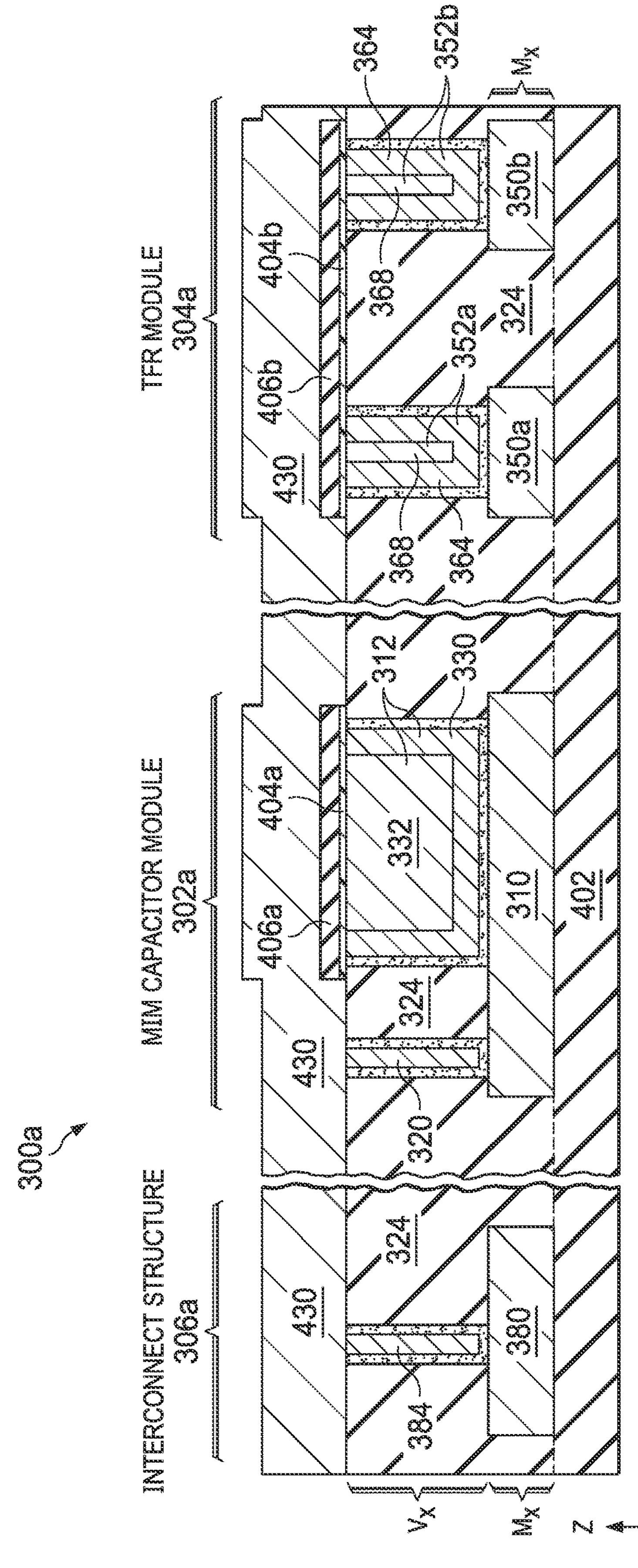

300a
INTERCONNECT STRUCTURE
306a
MIM CAPACITOR MODULE
302a
TFR MODULE
304a
430
324
380
384
320
310
402
406a
404a
332
312
330
406b
404b
368
364
352a
352b
350a
350b
V_X
M_X
Z
X

FIG. 11

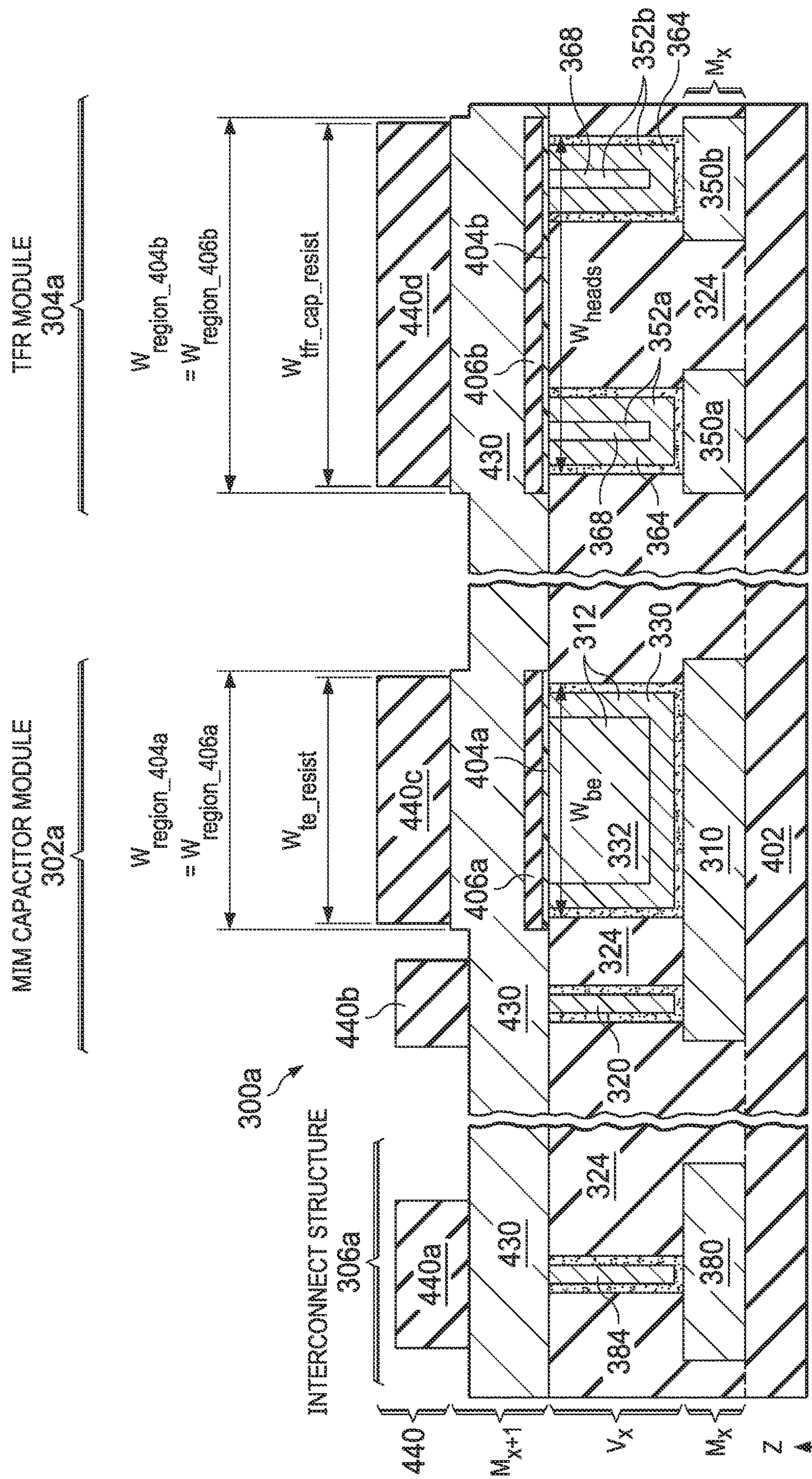

TFR MODULE
304a
MIM CAPACITOR MODULE
302a
INTERCONNECT STRUCTURE
306a
300a
$W_{region\_404b} = W_{region\_406b}$
$W_{tfr\_cap\_resist}$
$W_{region\_404a} = W_{region\_406a}$
W_{te_resist}
W_{heads}
W_{be}
368
352b
364
M_X
350b
404b
440d
324
352a
406b
430
350a
312
330
404a
440c
406a
332
310
402
440b
320
440a
380
384
440
M_{X+1}
V_X
Z
X

FIG. 12

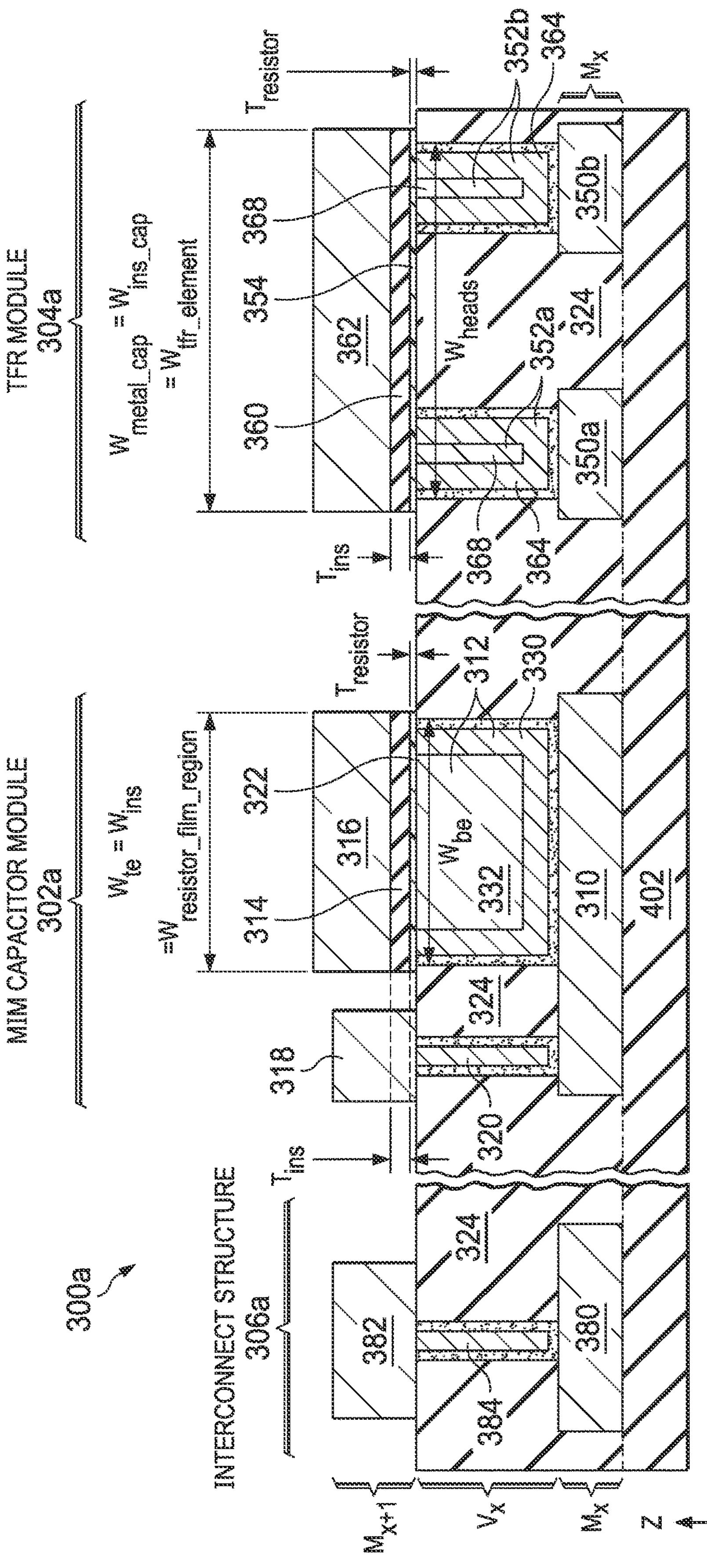
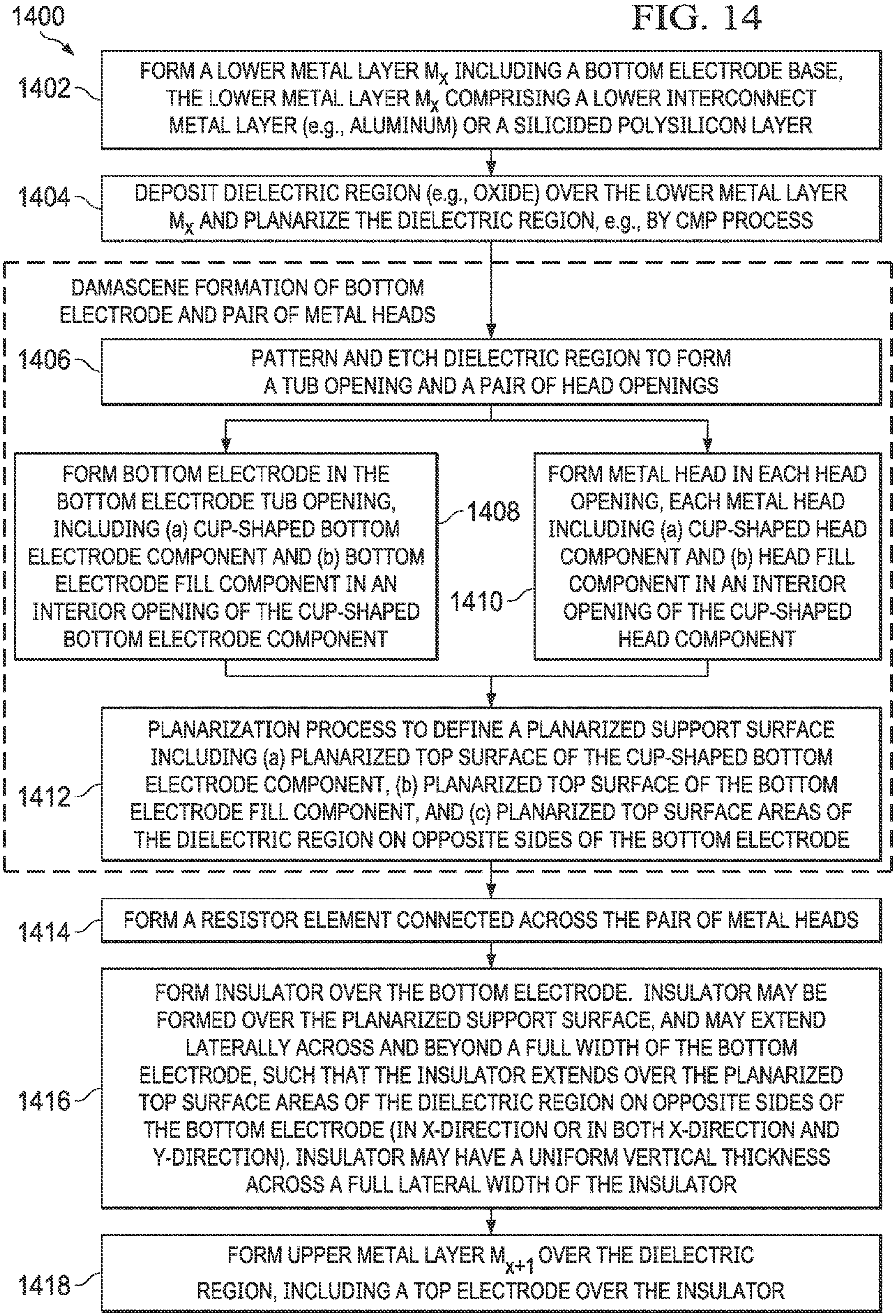

300a
INTERCONNECT STRUCTURE
306a
MIM CAPACITOR MODULE
302a
TFR MODULE
304a
$W_{te} = W_{ins}$
$= W_{resistor\_film\_region}$
$W_{metal\_cap} = W_{ins\_cap}$
$= W_{tfr\_element}$
$T_{resistor}$
T_{ins}
W_{be}
W_{heads}
M_{x+1}
V_x
M_x
Z
X
310
312
314
316
318
320
322
324
330
332
350a
350b
352a
352b
354
360
362
364
368
380
382
384
402

FIG. 13

FIG. 14


1400
1402
FORM A LOWER METAL LAYER M_X INCLUDING A BOTTOM ELECTRODE BASE, THE LOWER METAL LAYER M_X COMPRISING A LOWER INTERCONNECT METAL LAYER (e.g., ALUMINUM) OR A SILICIDED POLYSILICON LAYER
1404
DEPOSIT DIELECTRIC REGION (e.g., OXIDE) OVER THE LOWER METAL LAYER M_X AND PLANARIZE THE DIELECTRIC REGION, e.g., BY CMP PROCESS
DAMASCENE FORMATION OF BOTTOM ELECTRODE AND PAIR OF METAL HEADS
1406
PATTERN AND ETCH DIELECTRIC REGION TO FORM A TUB OPENING AND A PAIR OF HEAD OPENINGS
FORM BOTTOM ELECTRODE IN THE BOTTOM ELECTRODE TUB OPENING, INCLUDING (a) CUP-SHAPED BOTTOM ELECTRODE COMPONENT AND (b) BOTTOM ELECTRODE FILL COMPONENT IN AN INTERIOR OPENING OF THE CUP-SHAPED BOTTOM ELECTRODE COMPONENT
1408
1410
FORM METAL HEAD IN EACH HEAD OPENING, EACH METAL HEAD INCLUDING (a) CUP-SHAPED HEAD COMPONENT AND (b) HEAD FILL COMPONENT IN AN INTERIOR OPENING OF THE CUP-SHAPED HEAD COMPONENT
1412
PLANARIZATION PROCESS TO DEFINE A PLANARIZED SUPPORT SURFACE INCLUDING (a) PLANARIZED TOP SURFACE OF THE CUP-SHAPED BOTTOM ELECTRODE COMPONENT, (b) PLANARIZED TOP SURFACE OF THE BOTTOM ELECTRODE FILL COMPONENT, AND (c) PLANARIZED TOP SURFACE AREAS OF THE DIELECTRIC REGION ON OPPOSITE SIDES OF THE BOTTOM ELECTRODE
1414
FORM A RESISTOR ELEMENT CONNECTED ACROSS THE PAIR OF METAL HEADS
1416
FORM INSULATOR OVER THE BOTTOM ELECTRODE. INSULATOR MAY BE FORMED OVER THE PLANARIZED SUPPORT SURFACE, AND MAY EXTEND LATERALLY ACROSS AND BEYOND A FULL WIDTH OF THE BOTTOM ELECTRODE, SUCH THAT THE INSULATOR EXTENDS OVER THE PLANARIZED TOP SURFACE AREAS OF THE DIELECTRIC REGION ON OPPOSITE SIDES OF THE BOTTOM ELECTRODE (IN X-DIRECTION OR IN BOTH X-DIRECTION AND Y-DIRECTION). INSULATOR MAY HAVE A UNIFORM VERTICAL THICKNESS ACROSS A FULL LATERAL WIDTH OF THE INSULATOR
1418
FORM UPPER METAL LAYER M_{X+1} OVER THE DIELECTRIC REGION, INCLUDING A TOP ELECTRODE OVER THE INSULATOR

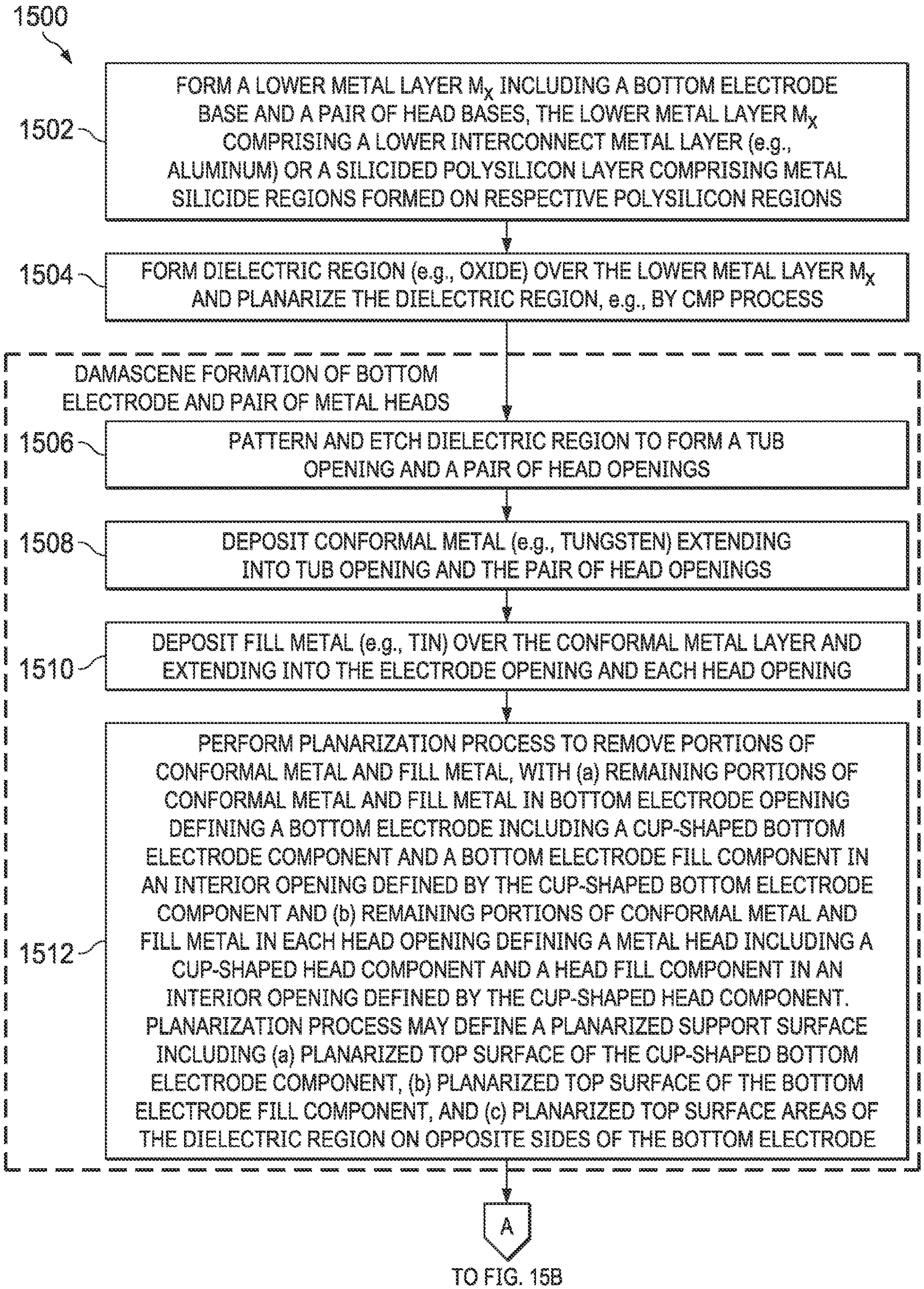

1500
1502
FORM A LOWER METAL LAYER M_X INCLUDING A BOTTOM ELECTRODE BASE AND A PAIR OF HEAD BASES, THE LOWER METAL LAYER M_X COMPRISING A LOWER INTERCONNECT METAL LAYER (e.g., ALUMINUM) OR A SILICIDED POLYSILICON LAYER COMPRISING METAL SILICIDE REGIONS FORMED ON RESPECTIVE POLYSILICON REGIONS
1504
FORM DIELECTRIC REGION (e.g., OXIDE) OVER THE LOWER METAL LAYER M_X AND PLANARIZE THE DIELECTRIC REGION, e.g., BY CMP PROCESS
DAMASCENE FORMATION OF BOTTOM ELECTRODE AND PAIR OF METAL HEADS
1506
PATTERN AND ETCH DIELECTRIC REGION TO FORM A TUB OPENING AND A PAIR OF HEAD OPENINGS
1508
DEPOSIT CONFORMAL METAL (e.g., TUNGSTEN) EXTENDING INTO TUB OPENING AND THE PAIR OF HEAD OPENINGS
1510
DEPOSIT FILL METAL (e.g., TIN) OVER THE CONFORMAL METAL LAYER AND EXTENDING INTO THE ELECTRODE OPENING AND EACH HEAD OPENING
1512
PERFORM PLANARIZATION PROCESS TO REMOVE PORTIONS OF CONFORMAL METAL AND FILL METAL, WITH (a) REMAINING PORTIONS OF CONFORMAL METAL AND FILL METAL IN BOTTOM ELECTRODE OPENING DEFINING A BOTTOM ELECTRODE INCLUDING A CUP-SHAPED BOTTOM ELECTRODE COMPONENT AND A BOTTOM ELECTRODE FILL COMPONENT IN AN INTERIOR OPENING DEFINED BY THE CUP-SHAPED BOTTOM ELECTRODE COMPONENT AND (b) REMAINING PORTIONS OF CONFORMAL METAL AND FILL METAL IN EACH HEAD OPENING DEFINING A METAL HEAD INCLUDING A CUP-SHAPED HEAD COMPONENT AND A HEAD FILL COMPONENT IN AN INTERIOR OPENING DEFINED BY THE CUP-SHAPED HEAD COMPONENT. PLANARIZATION PROCESS MAY DEFINE A PLANARIZED SUPPORT SURFACE INCLUDING (a) PLANARIZED TOP SURFACE OF THE CUP-SHAPED BOTTOM ELECTRODE COMPONENT, (b) PLANARIZED TOP SURFACE OF THE BOTTOM ELECTRODE FILL COMPONENT, AND (c) PLANARIZED TOP SURFACE AREAS OF THE DIELECTRIC REGION ON OPPOSITE SIDES OF THE BOTTOM ELECTRODE
A
TO FIG. 15B

FIG. 15A

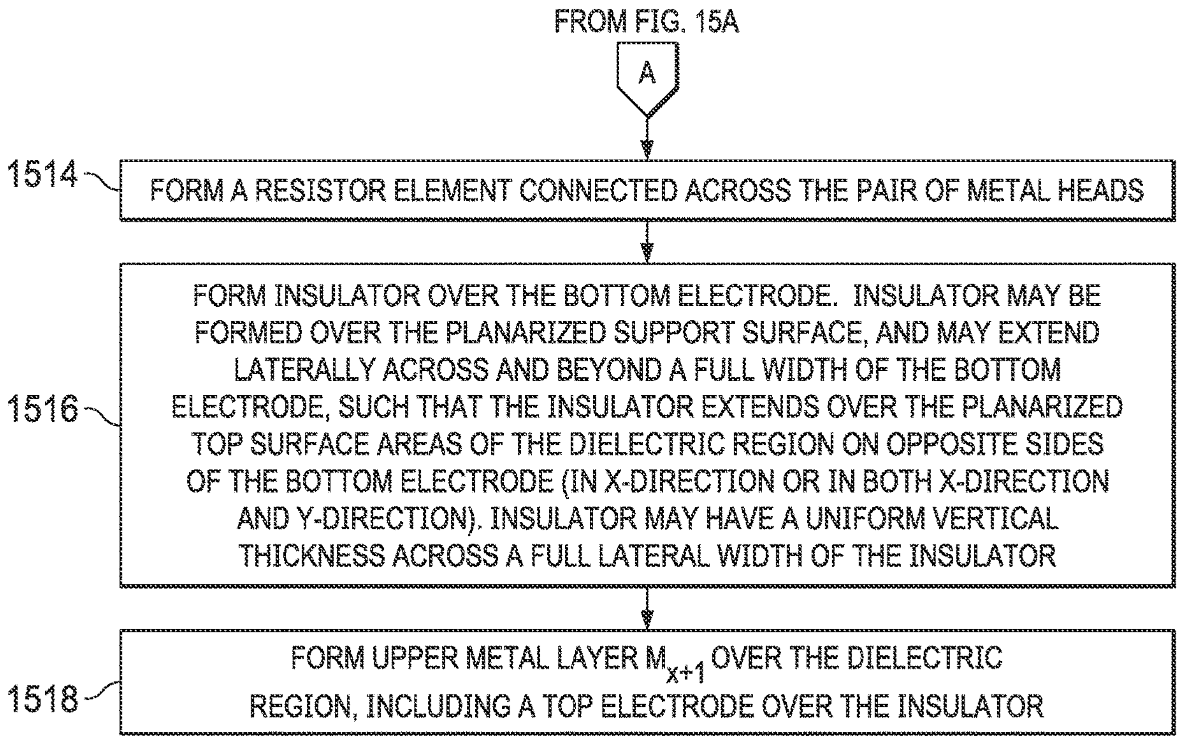

FROM FIG. 15A
A
1514
FORM A RESISTOR ELEMENT CONNECTED ACROSS THE PAIR OF METAL HEADS
1516
FORM INSULATOR OVER THE BOTTOM ELECTRODE. INSULATOR MAY BE FORMED OVER THE PLANARIZED SUPPORT SURFACE, AND MAY EXTEND LATERALLY ACROSS AND BEYOND A FULL WIDTH OF THE BOTTOM ELECTRODE, SUCH THAT THE INSULATOR EXTENDS OVER THE PLANARIZED TOP SURFACE AREAS OF THE DIELECTRIC REGION ON OPPOSITE SIDES OF THE BOTTOM ELECTRODE (IN X-DIRECTION OR IN BOTH X-DIRECTION AND Y-DIRECTION). INSULATOR MAY HAVE A UNIFORM VERTICAL THICKNESS ACROSS A FULL LATERAL WIDTH OF THE INSULATOR
1518
FORM UPPER METAL LAYER M_{x+1} OVER THE DIELECTRIC REGION, INCLUDING A TOP ELECTRODE OVER THE INSULATOR

FIG. 15B

INTEGRATED CIRCUIT STRUCTURE INCLUDING A METAL-INSULATOR-METAL (MIM) CAPACITOR MODULE AND A THIN-FILM RESISTOR (TFR) MODULE

RELATED PATENT APPLICATION

This application claims priority to commonly owned U.S. Provisional Patent Application No. 63/244,366 filed Sep. 15, 2021, the entire contents of which are hereby incorporated by reference for all purposes.

TECHNICAL FIELD

The present disclosure relates to analog components formed in integrated circuit devices, and more particularly to an integrated circuit structure including a metal-insulator-metal (MIM) capacitor module and a thin-film resistor (TFR) module.

BACKGROUND

Capacitors and resistors formed monolithically in integrated circuit devices are referred to as integrated capacitors and resistors, respectively. Integrated capacitors and resistors are common elements in many integrated circuit devices. For example, various analog, mixed signal, and RF-CMOS (radio-frequency complimentary metal-oxide-semiconductor) integrated circuit devices use integrated capacitors and resistors, either separately or in combination with each other. Integrated capacitors and resistors may offer various advantages over discrete counterparts (i.e., off-chip capacitors and resistors). For example, as compared with typical discrete (off-chip) capacitors and resistors, integrated capacitors and resistors may often be produced at lower cost. Additionally, system-on-chip devices including integrated capacitors and resistors may have a reduced pin count (which may provide improved ease-of-use and form factor), and may exhibit a reduced parasitic capacitance.

MIM capacitor modules are typically constructed between two interconnect metal layers (e.g., aluminum layers), referred to as metal layers M_x and M_{x+1}. For example, a MIM capacitor module may be formed using an existing metal layer M_x as the bottom electrode (bottom plate), constructing an insulator and a top electrode (top plate) over the bottom electrode, and connecting an overlying metal layer M_{x+1} to the top and bottom electrodes by respective vias. The top electrode formed between the two metal layers M_x and M_{x+1} may be formed from a different metal than the metal layers M_x and M_{x+1}. For example, the metal layers M_x and M_{x+1} may be formed from aluminum, whereas the top electrode may be formed from titanium/titanium nitride (Ti/TiN), tantalum/tantalum nitride (Ta/TaN), or tungsten (W), for example.

The top electrode typically has a higher resistance than the bottom electrode, for example because the top electrode may be limited by thickness constraints and the material of choice, thus limiting the performance of conventional MIM capacitor modules. MIM capacitor modules typically have very narrow process margins, particularly for a metal etch used to form the top electrode.

In addition, for MIM capacitor modules formed in aluminum interconnect (i.e., where metal layers M_x and M_{x+1} comprise aluminum interconnect layers), the aluminum bottom electrode may be susceptible to hillock formation at a top side of the bottom electrode, e.g., resulting from high-temperature processing of aluminum, a low-melting-point metal. Hillocks formed on the bottom electrode may negatively or unpredictably affect the breakdown voltage of the MIM capacitor module.

FIGS. **1**A-**1**F show cross-sectional side views of an example prior art process for forming an MIM capacitor module **10**. As shown in FIG. **1**A, a metal interconnect layer M_x is formed over a dielectric region, e.g., an inter-metal dielectric (IMD) layer indicated as IMD_x. Metal layer M_x may be formed from aluminum or other suitable metal.

Next, as shown in FIG. **1**B, an insulator layer **100** is deposited, followed by a top electrode layer **102** from which a top electrode (top plate) of the MIM capacitor module **10** is formed, as shown in FIG. **1**C discussed below. The insulator layer **100** may comprise silicon nitride (Si_3N_4, also referred to more simply as SiN) with a thickness T_{ins} of about 500 Å, which may be deposited by a plasma enhanced chemical vapor deposition (PECVD) process. The top electrode layer **102** may comprise titanium nitride (TiN), tantalum (Ta), tantalum nitride (TaN), or other suitable metal with thickness T_{te} of about 2000 Å. The thickness T_{te} of the top electrode layer **102**, which defines the thickness of the capacitor top electrode **108** (see FIG. **1**C), is typically limited by the thickness of a subsequently formed IMD layer IMD_{x+1} (shown in FIG. **1**F discussed below). In addition, an etch margin of the top electrode etch (see FIG. **1**C, discussed below), and a process margin of a subsequent chemical mechanical planarization (CMP), for example, may contribute to the limitation on the thickness T_{te} of the top electrode layer **102**. For a typical IMD_{x+1} layer thickness of 8000 Å, the thickness T_{te} of the top electrode **102** may be limited to about 3000 Å or less. This limited thickness T_{te} of the top electrode layer **102** (and thus capacitor top electrode **108**) can result in a high series resistance and low quality factor (Q-factor) for certain applications, e.g., radio frequency (RF) applications.

After the insulator layer **100** and top electrode layer **102** are deposited, information printed in the wafer scribe region of the underlying silicon substrate may be very difficult to read (through top electrode layer **102**, the insulator layer **100**, and underlying metal layer M_x), which can cause manufacturing problems. For example, the wafer lot number and/or wafer number printed in the wafer scribe region may be difficult to read, which may cause various problems, e.g., inability to run controlled experiments with designated wafers split among different process conditions or record wafer activities (e.g., scrap or incident events) tied to wafer number.

Next, as shown in FIG. **1**C, a photoresist layer is deposited and patterned to form a first photomask **106** over the top electrode layer **102**, and an etch is performed to define the MIM capacitor top electrode **108** from the top electrode layer **102**, wherein the portion of the insulator layer **100** below the MIM capacitor top electrode **108** defines the MIM capacitor insulator, indicated at **100**a. The top electrode etch typically has a very small process margin. In particular, the etch is designed to stop approximately halfway through the thickness T_{ins} of the insulator layer **100**.

The precise depth of the top electrode etch, which defines the thickness T_{ins_etched} of the remaining insulator layer **100** outside the footprint of the top electrode **108**, indicated as etched insulator layer region **100**b, is typically sensitive to specific process parameters, such as silicon nitride deposition thickness and non-uniformity, and etch uniformity and selectivity, for example. If the etch is insufficiently deep (such that T_{ins_etched} of the etched insulator layer region **100**b is too thick), the top electrode layer **102** may not be completely removed in some areas on the wafer (e.g., due to non-uniformity of the etch across the wafer), thus leaving metal residue or stringers on the wafer. These metal residue or stringers may cause incomplete etching in a subsequent metal etch step, create metal shorts, and lead to device yield loss or reliability failure.

On the other hand, if the top electrode etch is too deep (such that T_{ins_etched} of the etched insulator layer region **100***b* is too thin), the resulting MIM capacitor module **10** may have an unsuitably low breakdown voltage, in particular due to the corner "C" formed in the insulator layer **100** by the top electrode etch discussed above, as discussed below with respect to FIG. **1**F.

Thus, the effective margin for the thickness T_{ins_etched} of the etched insulator layer region **100***b* may be very small, thereby defining small process margins for the top electrode etch. For example, for a deposited silicon nitride insulator layer having a thickness T_{ins} of **500**A, an effective target thickness T_{ins_etched} of the etched insulator layer region **100***b* may have a very small margin, for example between 230-270 Å.

Next, as shown in FIG. **1**D, remaining portions of the first photomask **106** are removed and a new photoresist layer is deposited and patterned to form a second photomask **112** over the top electrode **108** and extending beyond the top electrode **108** in a first lateral direction.

Next, as shown in FIG. **1**E, a metal etch is performed to define the MIM capacitor bottom electrode **116**.

Finally, as shown in FIG. **1**F, construction of the MIM capacitor module **10** is completed by removing the second photomask **112**, forming an IMD layer (IMD_{x+1}), forming vias **120** respectively contacting the top electrode **108** and bottom electrode **116**, and forming a top electrode contact **122** and a bottom electrode contact **124** in a metal layer M_{x+1} over the IMD_{x+1} layer.

The prior art MIM capacitor module **10** may suffer from various shortcomings. For example, as noted above, the thickness T_{te} of the top electrode **108** may be limited due to a vertical spacing limitation between metal layers M_x and M_{x+1}, represented by the thickness T_{IMDx+1} of the IMD_{x+1} region. The limited thickness T_{te} of the top electrode **108** may result in high serial resistance unsuitable for certain applications (e.g., RF applications).

In addition, the MIM capacitor module **10** may have a low and/or unpredictable breakdown voltage. For example, the capacitor breakdown voltage may be very sensitive to the thickness T_{ins_etched} of the etched insulator layer region **100***b*, particularly at the corner C below the lateral edge of the top electrode **108**, as discussed above. In addition, the capacitor breakdown voltage may also be sensitive to hillocks "H" formed on the capacitor bottom electrode **116**. Hillock formation may be very difficult to control in the fabrication process discussed above. For example, hillocks H may form on the bottom electrode **116** as a result of various heated process steps during and after the capacitor module fabrication, including heat treatment steps and/or heated aluminum deposition steps (e.g., performed at 400° C.). These hillocks H may create an uncontrolled low breakdown voltage of the capacitor module **10**.

In addition, as noted above, the deposition of the various material layers (lower metal layer, insulation layer, and top electrode layer) over the wafer scribe region may hinder the ability to read information printed in the wafer scribe region (e.g., wafer number and lot number), which may complicate the manufacturing process.

Turning now to integrated resistors, one common type is the thin-film resistor (TFR) module, which includes a pair of metal heads connected by a resistor element, or TFR film.

FIG. **2** shows a cross-sectional view of two example TFR modules **200**A and **200**B implemented using conventional techniques. The fabrication of a conventional TFR module **200**A or **200**B typically requires three added mask layers, with reference to a background fabrication process for the relevant IC device. In particular, a first added mask layer may be used to create metal heads **202**A and **202**B, a second added mask layer may be used to create a resistor element **204**, and a third added mask layer may be used to create TFR vias **206**A and **206**B. As shown, the resistor element **204** of TFR module **200**A is formed across the top of the metal heads **202**A and **202**B, while the resistor element **204** of TFR module **200**B is formed across the bottom of the metal heads **202**A and **202**B, but each design typically uses three added mask layers.

TFR modules and MIM capacitor modules are often expensive to construct. For example, as discussed above, the process for forming an MIM capacitor module or TFR module often includes multiple additional mask layer to the background IC fabrication process. Also, MIM capacitor modules and TFR modules are typically constructed independent of each other, further compounding the number of additional mask layers needed to form both types of devices in an integrated circuit device.

There is a need to build integrated capacitors and resistors, in particular to construct MIM capacitor modules and TFR modules together, efficiently and at reduced cost as compared with conventional processes.

SUMMARY

The present disclosure provides an integrated circuit (IC) structure including both an MIM capacitor module and a TFR module formed concurrently in a dielectric region between two metal layers in the IC structure, e.g., an inter-metal dielectric (IMD) region or a pre-metal dielectric (PMD) region. In some examples, the IC structure also includes at least one interconnect structure formed in the dielectric region between the two metal layers in the IC structure, concurrently with the MIM capacitor module and the TFR module.

As used herein, a structure formed "between" two metal layers (e.g., an MIM capacitor module, TFR module, and/or interconnect structure formed between two layers, as disclosed herein) refers to a structure including (a) at least one component formed in a dielectric region between two metal layers and (b) at least one component formed in one or both of the two metal layers.

One aspect provides an integrated circuit structure including an MIM capacitor module and a TFR module. The MIM capacitor module includes a bottom electrode base formed in a lower metal layer, a bottom electrode formed in a dielectric region between the lower metal layer and an upper metal layer, an insulator formed over the bottom electrode, and a top electrode formed in the upper metal layer over the insulator. The bottom electrode includes a cup-shaped bottom electrode component, and a bottom electrode fill component formed in an interior opening defined by the cup-shaped bottom electrode component. The TFR module includes a pair of metal heads formed in the dielectric region between the lower metal layer and the upper metal layer, and a resistor element connected across the pair of metal heads. Each metal head of the pair of metal heads includes a cup-shaped head component, and a head fill component formed in an interior opening defined by the cup-shaped head component.

In one example, the TFR module includes a pair of head bases formed in the lower metal layer, wherein each metal head is conductively connected to a respective head base.

In one example, the cup-shaped head component of each metal head and the cup-shaped bottom electrode component are formed from a conformal metal, and the head fill component of each metal head and the bottom electrode fill component are formed from a fill metal different than the conformal metal. In one example, the conformal metal comprises tungsten, and the fill metal comprises titanium nitride.

In one example, the insulator has a uniform vertical thickness across a full lateral width of the insulator.

In one example, the insulator is formed over a planarized support surface including a planarized top surface of the cup-shaped bottom electrode component and a planarized top surface of the bottom electrode fill component.

In one example, the insulator is formed on a planarized support surface including (a) a planarized top surface of the cup-shaped bottom electrode component, (b) a planarized top surface of the bottom electrode fill component, and (c) planarized top surface areas of the dielectric region on opposite sides of the bottom electrode, and the insulator extends laterally across and beyond a full width of the bottom electrode, such that the insulator extends over the planarized top surface areas of the dielectric region on opposite sides of the bottom electrode.

In one example, the thin-film resistor module includes a resistor element insulator cap element formed on the resistor element, and the resistor element insulator cap element is formed from a common insulator material as the insulator.

In one example, the lower metal layer comprises a lower interconnect layer, and the upper metal layer comprises an upper interconnect layer. In another example, the lower metal layer comprises a silicided polysilicon layer, wherein the bottom electrode base formed in the lower metal layer comprises a metal silicide region formed on a polysilicon region, and the upper metal layer comprises a first metal interconnect layer.

In one example, the metal-insulator-metal capacitor module includes a bottom electrode connection element formed in the upper metal layer, and a bottom electrode contact formed in the dielectric region between the lower metal layer and the upper metal layer. The bottom electrode contact provides a conductive connection between the bottom electrode connection element formed in the upper metal layer and the bottom electrode base formed in the lower metal layer.

In some examples, the integrated circuit structure includes an interconnect structure including a lower interconnect element formed in the lower metal layer, and an upper interconnect contact formed in the upper metal layer and connected to the lower interconnect element by at least one interconnect via. In one example, the at least one interconnect via and the cup-shaped bottom electrode component are formed from a common conformal metal.

Another aspect provides a method of forming an integrated circuit structure. The method includes forming a lower metal layer including a bottom electrode base, depositing a dielectric region over the lower metal layer, patterning and etching the dielectric region to form a bottom electrode tub opening and a pair of head tub openings, and forming a bottom electrode in the bottom electrode tub opening, wherein the bottom electrode includes a cup-shaped bottom electrode component and a bottom electrode fill component in an interior opening defined by the cup-shaped bottom electrode component. The method also includes forming a metal head in each of the pair of head tub openings, wherein each metal head includes a cup-shaped head component and a head fill component in an interior opening defined by the cup-shaped head component. The method also includes forming an insulator over the bottom electrode, forming a resistor element connected across the pair of metal heads, and forming an upper metal layer including a top electrode over the insulator.

In one example, the method includes performing a planarization process to define a planarized support surface including a planarized top surface of the cup-shaped bottom electrode component and a planarized top surface of the bottom electrode fill component, and forming the insulator over the planarized support surface.

In one example, the method includes forming the insulator with a uniform vertical thickness across a full lateral width of the insulator.

In one example, the method includes performing a planarization process to define a planarized support surface including (a) a planarized top surface of the cup-shaped bottom electrode component, (b) a planarized top surface of the bottom electrode fill component, and (c) planarized top surface areas of the dielectric region on opposite sides of the bottom electrode. The insulator extends laterally across and beyond a full width of the bottom electrode, such that the insulator extends over the planarized top surface areas of the dielectric region on opposite sides of the bottom electrode.

Another aspect provides a method of forming an integrated circuit structure. The method includes forming a lower metal layer including a bottom electrode base and a pair of head bases, forming a dielectric region over the lower metal layer, forming openings in the dielectric region, including (a) a bottom electrode opening and (b) a pair of head openings, depositing a conformal metal extending into the bottom electrode opening and into each head opening, and depositing a fill metal over the conformal metal layer and extending into the electrode opening and into each head opening. The method also includes performing a planarization process to remove portions of both the conformal metal and the fill metal, wherein (a) remaining portions of the conformal metal and fill metal in the bottom electrode opening define a bottom electrode including (i) a cup-shaped bottom electrode component and (ii) a bottom electrode fill component in an interior opening defined by the cup-shaped bottom electrode component, and (b) remaining portions of the conformal metal and fill metal in each of the pair of head openings define a pair of metal heads, each metal head including (i) a cup-shaped head component and (ii) a head fill component in an interior opening defined by the cup-shaped head component. The method also includes forming an insulator over the bottom electrode, forming a resistor element spanning the pair of metal heads, and forming an upper metal layer including a top electrode over the insulator.

In one example, the method includes forming the insulator with a uniform vertical thickness across a full lateral width of the insulator.

In one example, the planarization process defines a planarized support surface including a planarized top surface of the cup-shaped bottom electrode component and a planarized top surface of the bottom electrode fill component, and the insulator is formed on the planarized support surface.

In one example, the planarization process defines a planarized support surface including (a) a planarized top surface of the cup-shaped bottom electrode component, (b) a planarized top surface of the bottom electrode fill component, and (c) planarized top surface areas of the dielectric region on opposite sides of the bottom electrode. The insulator extends laterally across and beyond a full width of the bottom electrode, such that the insulator extends over the planarized top surface areas of the dielectric region on opposite sides of the bottom electrode.

In one example, the method includes forming an insulator layer over the bottom electrode and over the pair of metal heads, and etching the insulator layer to form (a) the insulator over the bottom electrode and (b) an insulator cap element over the resistor element.

In one example, forming the upper metal layer includes forming (a) the top electrode over the insulator and (b) a metal cap element over the insulator cap element.

In one example, the method includes depositing a resistor film over the bottom electrode and over the pair of metal heads, depositing an insulator layer over the resistor film, and after depositing an insulator layer over the resistor film, patterning and etching the resistor film and the insulator layer. Remaining portions of the resistor film define (a) a resistor film region over the bottom electrode and (b) the resistor element spanning the pair of metal heads, and remaining portions of the insulator layer define (a) the insulator over the bottom electrode and (b) an insulator cap element over the resistor element.

In one example, the lower metal layer comprises a lower interconnect layer, and the upper metal layer comprises an upper interconnect layer.

In one example, the lower metal layer comprises a silicided polysilicon layer, wherein each of (a) the bottom electrode base and (b) each head base comprises a metal silicide region formed on a respective polysilicon region, and the upper metal layer comprises a first metal interconnect layer.

In one example, the method includes forming an interconnect via opening in the dielectric region, and depositing the conformal metal into the interconnect via opening, wherein a portion of the conformal metal in the interconnect via opening after the planarization process defines an interconnect via, and wherein the interconnect via conductively connects a lower interconnect element formed in the lower metal layer to an upper interconnect element formed in the upper metal layer.

BRIEF DESCRIPTION OF THE DRAWINGS

Example aspects of the present disclosure are described below in conjunction with the figures, in which:

FIGS. **1A-1F** are cross-sectional side views showing an example prior art process for forming an example MIM capacitor module;

FIG. **2** is a cross-sectional view of two example prior art thin-film resistors (TFR);

FIG. **3A** is a cross-sectional side view an example integrated structure including an example MIM capacitor module, TFR module, and interconnect structure formed between two metal interconnect layers in an integrated circuit structure, according to one example;

FIG. **3B** is a cross-sectional side view an example integrated structure including an example MIM capacitor module, TFR module, and interconnect structure formed between a silicided polysilicon layer and a first metal interconnect layer in an integrated circuit structure, according to one example;

FIGS. **4-13** are cross-sectional side views showing an example process of forming an integrated circuit structure including the example MIM capacitor module, TFR module and interconnect structure shown in FIG. **3A** along with an example interconnect structure;

FIG. **14** is a flowchart showing an example method of forming an example integrated circuit structure including an example MIM capacitor module and an example TFR module, according to one example; and

FIGS. **15A-15B** illustrate a flowchart showing another example method of forming an example integrated circuit structure including an example MIM capacitor module and an example TFR module, according to another example.

It should be understood the reference number for any illustrated element that appears in multiple different figures has the same meaning across the multiple figures, and the mention or discussion herein of any illustrated element in the context of any particular figure also applies to each other figure, if any, in which that same illustrated element is shown.

DETAILED DESCRIPTION

The present disclosure provides an integrated circuit (IC) structure including both an MIM capacitor module and a TFR module formed in a dielectric region between two metal layers in the IC structure, e.g., an inter-metal dielectric (IMD) region or a pre-metal dielectric (PMD) region.

The MIM capacitor module includes (a) a bottom electrode base formed in a lower metal layer M_x, (b) a bottom electrode conductively coupled to the bottom electrode base, (c) an insulator formed over the bottom electrode, and (d) a top electrode formed in an upper metal layer M_{x+1} over the insulator. The MIM capacitor module may also include (e) a bottom electrode connection element formed in the upper metal layer M_{x+1}, and (f) a bottom electrode contact conductively connecting the bottom electrode connection element to the bottom electrode base.

The TFR module includes (a) a pair of head bases formed in the lower metal layer M_x, (b) a pair of metal heads, each conductively connected to one of the head bases, and (c) a resistor element spanning across and conductively connected to the pair of metal heads. The TFR module may also include a resistor element insulator cap element over the resistor element, and a metal cap element over the resistor element insulator cap element, wherein the resistor element insulator cap element is formed from a common resistor film as the insulator of the MIM capacitor module and the metal cap element is formed in the same upper metal layer M_{x+1} as the top electrode and bottom electrode contact of the MIM capacitor module.

Thus, the bottom electrode of the MIM capacitor module and each metal head of the TFR module are formed between the lower metal layer M_x and upper metal layer M_{x+1}. In addition, the bottom electrode and each metal head may be formed as a multi-component structure using a damascene process. The bottom electrode of the MIM capacitor module may include a cup-shaped bottom electrode component and a bottom electrode fill component formed in an interior opening defined by the cup-shaped bottom electrode component. Similarly, each metal head of the TFR module may include a cup-shaped head component and a head fill component formed in an interior opening defined by the cup-shaped head component. The cup-shaped bottom electrode component of the bottom electrode, and the cup-shaped head component of each metal head of the TFR module, may be formed from a common conformal metal, e.g., tungsten. The bottom electrode fill component of the bottom electrode, and the head fill component of each metal head of the TFR module, may be formed from a fill metal, e.g., titanium nitride.

The insulator of the MIM capacitor module may be formed on a planarized support surface (e.g., including a planarized upper surface of the bottom electrode). The insulator may have a uniform thickness across a full lateral width of the insulator, which may provide an improved capacitor breakdown voltage as compared with certain conventional capacitor modules. In addition, by forming the bottom electrode from refractory metals, e.g., tungsten and titanium nitride, the top surface of the bottom electrode that interfaces the insulator may be free of hillocks that are common in certain conventional capacitors (e.g., capacitor modules using an aluminum bottom electrode), which may provide a higher and more consistent capacitor breakdown voltage as compared with such conventional capacitor modules. Also, the top electrode and the bottom electrode may each have a substantial thickness, e.g., a thickness of at least 4000 Å, which may provide improved performance in particular applications (e.g., RF applications) as compared with certain conventional capacitor modules having a thinner top electrode and/or bottom electrode.

As mentioned above, the MIM capacitor module and TFR module are formed between two metal layers, in particular the lower metal layer M_x (in which the bottom electrode base of the MIM capacitor, and the pair of head bases of the TFR module, are formed) and an upper metal layer M_{x+1} (in which the top electrode and bottom electrode connection element of the MIM capacitor module, and the metal cap element over the resistor element of the TFR module, are formed). As used herein, a "metal layer," for example in the context of the lower metal layer M_x and upper metal layer M_{x+1}, may comprise any metal or metalized layer or layers, including:

(a) a metal interconnect layer, e.g., comprising aluminum, copper, or other metal formed by a damascene process or deposited by a subtractive patterning process (e.g., deposition, patterning, and etching of a metal layer), or
(b) a silicided polysilicon layer including a number of silicided polysilicon structures (i.e., polysilicon structures having a metal silicide layer formed thereon), for example a silicided polysilicon gate of a metal-oxide-semiconductor field-effect transistor (MOSFET).

For example, as shown in FIG. **3**A, an MIM capacitor module and TFR module may be constructed between two adjacent metal interconnect layers M_x and M_{x+1} at any depth in an integrated circuit structure. As another example, as shown in FIG. **3**B, an MIM capacitor module and TFR module may be constructed between a silicided polysilicon layer and a first metal interconnect layer (commonly referred to as the metal-1 layer), wherein the silicided polysilicon layer defines the lower metal layer M_x where x=0 (i.e., M_0) and the first metal interconnect layer (Metal-1 layer) defines the upper metal layer M_{x+1} (i.e., M_1).

FIG. **3**A is a cross-sectional side view an example integrated circuit structure **300***a* including an example MIM capacitor module **302***a*, an example TFR module **304***a*, and (optionally) an example interconnect structure **306***a*, all formed between two metal interconnect layers M_x and M_{x+1}, according to some examples. In some examples, integrated circuit structure **300***a* includes the MIM capacitor module **302***a*, TFR module **304***a*, and interconnect structure **306***a*. In other examples, integrated circuit structure **300***a* includes the MIM capacitor module **302***a* and TFR module **304***a*, but not the interconnect structure **306***a*. Thus, the optional interconnect structure **306***a* is indicated in FIG. **3**A with a dashed line.

The example MIM capacitor module **300***a* includes (a) a bottom electrode base **310** formed in a lower metal interconnect layer M_x, (b) a bottom electrode **312** conductively coupled to the bottom electrode base **310**, (c) an insulator **314** formed over the bottom electrode **312**, and (d) a top electrode **316** formed in an upper metal interconnect layer M_{x+1} over the insulator **314**. A resistor film region **322** may be arranged between the insulator **314** and underlying bottom electrode **312**. The MIM capacitor module **302***a* also includes a bottom electrode connection element **318** formed in the upper metal interconnect layer M_{x+1}, and a bottom electrode contact **320** conductively connecting the bottom electrode connection element **318** to the bottom electrode base **310**.

The bottom electrode **312** is formed in a dielectric region **324** formed in a via layer V_x between the metal interconnect layer M_x and upper metal interconnect layer M_{x+1}. The bottom electrode **312** may be formed using a damascene process, as described below. In the illustrated example, the bottom electrode **312** includes (a) a cup-shaped bottom electrode component **330** formed in a tub opening **331** formed in the dielectric region **324** and (b) a bottom electrode fill component **332** formed in an interior opening **333** defined by the cup-shaped bottom electrode component **330**.

In some examples, the bottom electrode contact **320** is formed as a via, and may also be referred to as a bottom electrode via.

In some examples, the insulator **314** may be formed over a planarized support surface **340**. For example, as shown in FIG. **3**A, the insulator **314** may be formed on the resistor film region **322** formed on the planarized support surface **340**. The planarized support surface **340**, which may be formed by a chemical mechanical planarization (CMP) or other planarization process, includes (a) a planarized top surface **341** of the cup-shaped bottom electrode component **330**, (b) a planarized top surface **342** of the bottom electrode fill component **332**, and (c) a planarized top surface **347** of the dielectric region **324**, including top surface areas of the dielectric region **324** on opposite lateral sides (in the x-direction, or in both the x-direction and y-direction) of each of the bottom electrode **312**.

The planarized support surface **340** may free of metal hillocks, as the bottom electrode **312** is formed from refractory metals (e.g., tungsten and titanium nitride) which resist the formation of hillocks, unlike the aluminum bottom electrode of the example prior art MIM capacitor module **10** shown in FIG. **1**F discussed above. Forming the insulator **314** on a surface free of hillocks may provide a higher and more consistent capacitor breakdown voltage, e.g., as compared with a conventional capacitor module having a bottom electrode including hillocks.

The surface roughness of the planarized support surface **340** may depend on the specific process parameters implemented (e.g., CMP process parameters). In some examples, the planarized support surface **340** has a root-mean-square (RMS) surface roughness of less than 50 Å. In some examples, the planarized support surface **340** has an RMS surface roughness of less than 20 Å.

As mentioned above, insulator **314** may be formed on resistor film region **322** formed on the planarized support surface **340**. The lateral width W_{ins} of the insulator **314** in the x-direction may be coextensive with the lateral width W_{te} of the overlying top electrode **316**, as a result of an anisotropic metal etch extending through both the top electrode **316** and insulator **314**, e.g., as shown in FIG. **13** discussed below.

In some examples, e.g., as shown in FIG. **3**A, a lateral width W_{te} of the top electrode **316** and a corresponding lateral width W_{ins} of insulator **314** are greater than a lateral width W_{be} of the underlying bottom electrode **312** in the x-direction, such that the insulator **314** extends across and beyond the full lateral width W_{be} of the bottom electrode **312** and onto the planarized top surface areas **347** of dielectric region **324** on opposite sides of the bottom electrode **312** in the x-direction. In other words, the insulator **314** extends across the planarized top surface **341** of the cup-shaped bottom electrode component **330**, the planarized top surface **342** of the bottom electrode fill component **332**, and the planarized top surface areas **347** of the dielectric region **324**. In some examples, the insulator **314** extends laterally across the full lateral width W_{be} of the bottom electrode **312** in both the x-direction and y-direction, and additionally extends over portions of the dielectric region **324** on all lateral sides of the bottom electrode **312**, such that a perimeter of the insulator **314** surrounds a perimeter of the bottom electrode **312**, from a top view.

The insulator **314** has a uniform thickness T_{ins} across the full lateral width W_{ins} of the insulator **314**. For example, in some examples the thickness T_{ins} of the insulator **314** varies by less than 10% across the full lateral width W_{ins} of the insulator **314**. In some implementations the thickness T_{ins} of the insulator **314** varies by less than 5%, or even less than 1%, across the full lateral width W_{ins} of the insulator **314**.

This uniform thickness T_{ins} across the full lateral width W_{ins} of the insulator **314** may provide an increased and predictable breakdown voltage for the resulting MIM capacitor module **302***a*, e.g., as compared with a capacitor module having a partially etched thickness or otherwise varying thickness, e.g., the prior art MIM capacitor module **10** shown in FIG. **1**F and discussed above.

Further, by forming the bottom electrode **312** between two metal layers M_x and M_{x+1} the bottom electrode **312** may have a thickness T_{be} extending across the full thickness of the dielectric region **324** between the metal layers M_x and M_{x+1}. In addition, by forming the top electrode **316** from the metal interconnect layer M_{x+1}, the top electrode **316** may have a thickness T_{te} defined by the thickness of the metal interconnect layer M_{x+1}. Thus, the bottom electrode thickness T_{be} and top electrode thickness T_{te} may be sufficient to provide target performance characteristics for various applications (e.g., including RF applications), as compared with certain conventional capacitor modules having a thinner top electrode and/or bottom electrode. In some examples, the bottom electrode thickness T_{be} may be at least 4000 Å, and the top electrode thickness T_{te} may be at least 4000 Å.

Next, as shown in FIG. **3**A, the example TFR module **304***a* includes (a) a pair of head bases **350***a* and **350***b* formed in the lower metal interconnect layer M_x, (b) a pair of metal heads **352***a* and **352***b*, each conductively coupled to a respective head base **350***a*, **350***b*, and (c) a resistor element **354** connected across the pair of metal heads **352***a* and **352***b*. The example TFR module **304***a* also includes (a) an insulator cap element **360** (formed from an insulator layer used to form the insulator **314** of the MIM capacitor module **302***a*), and (*b*) a metal resistor cap element **362** formed in the upper metal interconnect layer M_{x+1} over the insulator cap element **360**.

Metal heads **352***a* and **352***b* are formed in the via layer V_x between the lower metal interconnect layer M_x and upper metal interconnect layer M_{x+1}, e.g., using a damascene process. For example, each metal head **352***a* and **352***b* may include (a) a cup-shaped head component **364** formed in a head opening **365** formed in the dielectric region **324** and (b) a head fill component **368** formed in an interior opening **369** defined by the cup-shaped head component **364**.

The resistor element **354** is formed on the polished, planarized support surface **340**. For example, as shown in FIG. 3A, the resistor element **354** may be formed on a portion of the planarized support surface **340** including (a) a planarized top surface **345** of each cup-shaped head component **364**, (b) a planarized top surface **346** of each head fill component **368**, and (c) a planarized top surface areas **347** of the dielectric region **324** laterally between and adjacent the pair of metal heads **352***a* and **352***b*.

The resistor element **354** of the TFR module **304***a* may be formed from a common resistor film as the resistor film region **322** formed between the insulator **314** and underlying bottom electrode **312** of the MIM capacitor module **302***a*, as described below. Further, the insulator cap element **360** of the TFR module **304***a* may be formed from a common insulator layer as the insulator **314** of the MIM capacitor module **302***a*, as described below.

The (optional) interconnect structure **306***a* includes a lower interconnect element **380** formed in the lower metal interconnect layer M_x, an upper interconnect contact **382** formed in the upper metal interconnect layer M_{x+1}, and an interconnect via **384** formed in the via layer V_x and conductively connected between the lower interconnect element **380** and upper interconnect contact **382**.

The cup-shaped bottom electrode component **330** of the bottom electrode **312**, the cup-shaped head component **364** of each metal head **352***a*, **352***b*, the bottom electrode contact **320**, and the interconnect via **384** may be formed from a conformal metal, e.g., tungsten. In some examples, the cup-shaped bottom electrode component **330**, the cup-shaped head components **364** of metal heads **352***a* and **352***b*, the bottom electrode contact **320**, and the interconnect via **384** may be formed simultaneously by deposition of tungsten or other conformal metal, as discussed below with reference to FIG. **6**. In some examples, an adhesive layer **326** (e.g., a TiN layer with a thickness in the range of 50-300 Å) is deposited prior to deposition of the conformal metal to improve adhesion between the conformal metal and surrounding areas of the dielectric region **324**, in particular in the tub opening **331** and head openings **365**.

The bottom electrode fill component **332** of the bottom electrode **312**, and the head fill component **368** of each metal head **352***a*, **352***b* may be formed from a "fill metal" that may be different than the conformal metal that forms the cup-shaped bottom electrode component **330**, each cup-shaped head component **364**, the bottom electrode contact **320**, and the interconnect via **384**. For example, the fill metal forming the bottom electrode fill component **332** and each head fill component **368** may comprise TiN or other refractory metal different than the conformal metal. In some examples, the bottom electrode fill component **332** and the head fill components **368** may be formed simultaneously by deposition of the fill metal, as discussed below with reference to FIG. **7**.

The example TFR module **304***a* shown in FIG. **3**A may include various advantages or improvements over certain conventional designs, e.g., the example prior art TFR designs shown in FIG. **2**. First, the construction of TFR module **304***a* may be simplified, as compared with conventional methods. For example, the TFR module **304***a* may be constructed using only a single added mask added to the relevant background IC fabrication process. In addition, the metal heads **352***a*, **352***b* of the TFR module **304***a* may be built concurrently with the bottom electrode **312** of the MIM capacitor module **302***a* and/or interconnect via **384**. The metal heads **352***a*, **352***b* may be formed with a relatively large contact areas and low resistance.

FIG. **3**B is a cross-sectional side view an example integrated circuit structure **300***b* including an example MIM capacitor module **302***b*, an example TFR module **304***b*, and (optionally) an example interconnect structure **306***b*, all formed between a silicided polysilicon layer and a first metal interconnect layer (commonly referred to as the metal-1 layer), according to one example. In this example, the silicided polysilicon layer defines a lower metal layer M_0 (i.e., lower metal layer M_x where x=0) and the first metal interconnect layer (Metal-1) defines an upper metal layer M_1.

Integrated circuit structure **300***b* is similar to integrated circuit structure **300***a* shown in FIG. **3**A and discussed above, with the exception that the illustrated structures of integrated circuit structure **300***b* (MIM capacitor module **302***b*, TFR module **304***b*, and optional interconnect structure **306***b*) are formed between the silicided polysilicon layer M_0 and first metal interconnect layer M_1, rather than between two metal interconnect layers M_x and M_{x+1}.

In some examples, integrated circuit structure **300***b* includes the MIM capacitor module **302***b*, TFR module **304***b*, and interconnect structure **306***b*. In other examples, integrated circuit structure **300***b* includes the MIM capacitor module **302***b* and TFR module **304***b*, but not the interconnect structure **306***b*. Thus, the optional interconnect structure **306***b* is indicated in FIG. **3**B with a dashed line.

As shown in FIG. **3**B, a bottom electrode base **310'** of the MIM capacitor module **300***b*, a pair of head bases **350***a'* and **350***b'* of the TFR module **304***b*, and a lower interconnect element **380'** of the interconnect structure **306***b* are each formed as a silicided polysilicon structure including a metal silicide layer formed on a respective polysilicon structure in the silicided polysilicon layer M_0. More particularly, bottom electrode base **310'** comprises a metal silicide layer **392***a* formed on a top surface of a polysilicon structure **390***a*; head base **350***a'* comprises a metal silicide layer **392***b* formed on a top surface of a polysilicon structure **390***b*; head base **350***b'* comprises a metal silicide layer **392***c* formed on a top surface of a polysilicon structure **390***c*; and lower interconnect element **380'** comprises a metal silicide layer **392***d* formed on a top surface of a polysilicon structure **390***d*. Metal silicide layers **392***a*-**392***d* may comprise any suitable metal silicide layer, for example titanium silicide (TiSi2), cobalt silicide (CoSi2), or nickel silicide (NiSi), having a thickness in the range of 100-300 Å or other suitable thickness.

FIGS. **4**-**13** are cross-sectional side views showing an example process for forming the example integrated circuit structure **300***a* shown in FIG. **3**A, including the example MIM capacitor module **302***a*, the example TFR module **304***a*, and the example interconnect structure **306***a*, according to one example.

First, as shown in FIG. **4**, a lower metal interconnect layer M_x is formed over a dielectric region **402**, e.g., a pre-metal dielectric (PMD) region or inter-metal dielectric (IMD) region. Metal layer M_x may be formed from aluminum, copper, or other suitable metal. Metal layer M_x may be deposited, patterned, and etched to form (a) the bottom electrode base **310** of the MIM capacitor module **302***a*, (b) the pair of head bases **350***a* and **350***b* of the TFR module **304***a*, and (c) the lower interconnect element **380** of the interconnect structure **306***a*. Alternatively, the bottom electrode base **310**, head bases **350***a* and **350***b*, and lower interconnect element **380** may be formed by a damascene process. Each of the bottom electrode base **310**, head bases **350***a* and **350***b*, and lower interconnect element **380** may comprise a wire or other laterally elongated structure (e.g., elongated in the y-axis direction), or a discrete pad (e.g., having a square, circular, or substantially square or circular shape in the x-y plane), or any other suitable shape and structure.

A resist strip may be performed to remove remaining portions of the photomask used for patterning the metal layer M_x. A dielectric region **324** may be formed over metal layer M_x, e.g., by performing an oxide deposition (e.g., using high density plasma (HDP) and PECVD processes) followed by a CMP process to planarize the oxide. Dielectric region **324** may be an inter-metal dielectric (IMD) region.

Next, as shown in FIG. **5**A (cross-sectional side view) and corresponding FIG. **5**B (top view), the dielectric region **324** is patterned and etched (e.g., using a plasma etch) to concurrently form (a) at least one interconnect via opening **410** for various interconnect structures, (b) a bottom electrode contact opening **412**, (c) the tub opening **331** for forming the bottom electrode **312** of the MIM capacitor module **302***a*, and (d) the pair of head openings **365** for forming the pair of metal heads **352***a* and **352***b* of the TFR module **304***a*.

The interconnect via opening **410** and bottom electrode contact opening **412** may each be formed as a narrow via opening with a lateral diameter or width W_{via}. In contrast, the tub opening **331** and each head opening **365** may have a substantially larger width (x-direction) and/or length (y-direction) than the narrow via openings **310** and **312**. The shape and dimensions of the tub opening **331** and head openings **365** may be selected based on various parameters, e.g., for effective manufacturing of the MIM capacitor module **302***a* and TFR module **304***a* (e.g., effective deposition of the conformal metal and fill metal (e.g., tungsten and TiN, respectively) into the openings **331** and **365**) and/or for desired performance characteristics of the resulting MIM capacitor module **302***a* and TFR module **304***a*. In some examples, the tub opening **331** and head openings **365** may each have a square or rectangular shape from a top view, e.g., as shown in FIG. **5**B. In other examples, the tub opening **331** may have a circular or oval shape from the top view.

As noted above, an x-direction width W_{tub} and/or y-direction length L_{tub} of the tub opening **331** may be substantially larger than the width W_{via} of each via opening. For example, in some embodiments, the width W_{tub} and/or length L_{tub} of the tub opening **331** is at least twice as large as the width W_{via} of each via opening. In particular embodiments, the width W_{tub} and/or length L_{tub} of the tub opening **331** is at least five times as large, ten times as large, or 20 times as large, as the width W_{via} of each via opening. In some examples, the width W_{via} of each via opening is in the range of 0.1-0.5 μm, whereas the width W_{tub} and length L_{tub} of the tub opening **331** are each the range of 1-100 μm.

In addition, an x-direction width W_{head} and/or y-direction length L_{head} of each head opening **365** may be substantially larger than the width W_{via} of each via opening. For example, in some embodiments, the width W_{head} and/or length L_{head} of the tub opening **365** is at least twice as large as the width W_{via} of each via opening. In particular embodiments, the width W_{head} and/or length L_{head} of each head opening **365** is at least five times as large, ten times as large, or 20 times as large as the width W_{via} of each via opening. In some examples, the width W_{via} of each via opening is in the range of 0.1-0.5 μm, whereas the width W_{head} and length L_{head} of each head opening **365** are each the range of 0.5-10 μm.

After the etch to create the openings **410**, **412**, **331**, and **365**, any remaining photoresist material may be removed by a resist strip.

Next, as shown in FIG. **6**, adhesive layer **326** (e.g., comprising titanium nitride, TiN) is deposited over the integrated circuit structure **300***a* and into the openings **410**, **412**, **331**, and **365**, e.g., with a thickness in the range of 50-300 Å, in one example. A conformal metal layer **420**, e.g., tungsten (W), is then deposited over adhesive layer **326**, e.g., by Chemical Vapor Deposition (CVD) deposition, with a thickness, in one example, in the range of 1000-5000 Å. As shown, the conformal metal layer **420** (*a*) fills the interconnect via opening **410** to form an interconnect via **384**, (b) fills the bottom electrode contact opening **412** to form the bottom electrode contact **320**, (c) partially fills the tub opening **331** to form the cup-shaped bottom electrode component **330** in the tub opening **331**, and (d) partially fills each head opening **365** to form the cup-shaped head component **364** in each head opening **365**. The cup-shaped bottom electrode component **330** defines an interior opening **333**, and each cup-shaped head component **364** defines an interior opening **369**.

The interconnect via **384**, bottom electrode contact **320**, cup-shaped bottom electrode component **330**, and cup-shaped head components **364** are further processed as discussed below, including a CMP process shown in FIG. **8**A, which defines a final form of each respective element **384**, **320**, and **330**, and **364**.

The deposited conformal metal layer **420** layer may have high tensile stresses, due to inherent material properties of the conformal metal, e.g., tungsten. As a result, a deposition thickness substantially above about 5000 Å (e.g., a thickness of 7000 Å) may result in a cracking or peeling of the conformal metal layer **420**, or a warping or breakage of the underlying silicon wafer (not shown), e.g., during a subsequent CMP process. However, these dimensions are not meant to be limiting in any way.

Next, as shown in FIG. **7**, a fill metal layer **430** is deposited over the integrated circuit structure **300** to fill the interior opening **333** of the cup-shaped bottom electrode component **330** and the interior opening **369** of each cup-shaped head component **364**. The fill metal layer **330** may deposited with sufficient thickness to fill the interior openings **333** and **369** at least to the top of the dielectric region **324**. The portion of the fill metal layer **430** filling interior opening **333** of the cup-shaped bottom electrode component **330** defines the bottom electrode fill component **332**, and the portion of the fill metal layer **430** filling the interior opening **369** of each cup-shaped head component **364** defines a respective head fill component **368**.

The fill metal layer **430** may be deposited by a reactive PVD or CVD process without generating hillocks, which may provide an increased breakdown voltage for the resulting MIM capacitor module **302***a*, as compared with prior art designs that enable or encourage hillock formation on the capacitor bottom electrode (and/or other capacitor components), e.g., as discussed above regarding the example prior art capacitor module **10** shown in FIG. **1**F.

In some examples, the fill metal layer **430** comprises titanium nitride (TiN) or other refractory metal (different from the conformal metal of the conformal metal layer **420**) that has inherent compressive stresses (e.g., for a layer thickness of less than 1 μm). The inherent compressive stresses of the fill metal layer **430** (e.g., titanium nitride layer) may counteract the inherent tensile stresses of the underlying conformal metal layer **420** (e.g., tungsten layer), to thereby reduce the risk of inter-layer peeling, silicon wafer breakage, or other mechanical failure. In another example, the fill metal layer **430** is formed of aluminum, which may provide reduced resistance for the resulting bottom electrode **312** and metal heads **352***a*, **352***b*, but may introduce the possibility of hillock formation.

Next, as shown in FIG. **8**A (cross-sectional side view) and corresponding FIG. **8**B (top view), a planarization process (e.g., CMP process) is performed to remove upper portions of the fill metal layer **430**, the conformal metal layer **420**, and adhesive layer **326**, and thereby define the final form of (a) the interconnect via **384**, (b) the bottom electrode contact **320**, (c) the cup-shaped bottom electrode component **330** and bottom electrode fill component **332** of the bottom electrode **312** of the MIM capacitor module **302***a*, and (d) the cup-shaped head component **364** and head fill component **368** of each metal head **352***a*, **352***b* of the TFR module **304***a*. The planarization process may be designed to stop on the dielectric region **324**.

The planarization process (e.g., CMP process) defines a polished, planarized support surface **340** that includes (a) the planarized top surface **341** of the cup-shaped bottom electrode component **330**, (b) the planarized top surface **342** of the bottom electrode fill component **332**, (c) the planarized top surface **345** of each cup-shaped head component **368**, (d) the planarized top surface **346** of each head fill component **368**, and (e) the planarized top surface **347** of the dielectric region **324**, including top surface areas of the dielectric region **324** on opposite lateral sides (in the x-direction, or in both the x-direction and y-direction) of each of the bottom electrode **312** and each metal head **352***a*, **352***b*.

The planarized support surface **340** provides a smooth, planarized surface free of hillocks. For example, the planarized top surfaces **341**, **342**, **345**, and **346** (components of the planarized support surface **340**) provide smooth, planarized surfaces of refractory metals used as the conformal metal layer **420** and fill metal layer **430** (e.g., tungsten and titanium nitride, as discussed above) that are free of hillocks, suitable for supporting the insulator **314** and resistor element **354**. As discussed below, the resistor element **354** is formed from a resistor film **404** formed directly on the planarized support surface **340**, whereas the insulator **314** is formed on a resistor film region **322** formed from the same resistor film **404**. As discussed below, the resistor film **404** is formed as a planar layer with a uniform thickness across the planarized support surface **340**; thus, the insulator **314** formed on the resistor film region **322** is also formed as a planar structure with a uniform thickness across the lateral width of the insulator **314** (in the x-direction, or both the x-direction and y-direction).

Next, as shown in FIG. **9**, a resistor film **404** is deposited over the structure **300**, followed by deposition of an insulator layer **406**. In some examples, the resistor film **404** may comprise SiCCr, SiCr, NiCr, TaN, or other suitable material, deposited with a thickness $T_{resistor}$, which in one example may be in the range of 100-1000 Å, for example by a physical vapor deposition (PVD) process, and having a sheet resistance, in one example, in the range of 200-1,000 ohms per square. The insulator layer **406** may comprise silicon nitride (Si_3N_4, κ~7) deposited with a thickness T_{ins}, in one example in the range of 400-600 Å, for example by a PECVD deposition process. In other examples, the insulator layer **406** may comprise silicon oxide (Sift) or a high-k dielectric material, e.g., Al_2O_3 (κ~10), Ta_2O_5 (κ~25), HfO_2 (κ~22), or ZrO_2 (κ~35), e.g., deposited by a PECVD deposition process or an ALD (atomic layer deposition) process.

A photoresist layer is then deposited and patterned to form a photomask **410** over the insulator layer **406**. The patterned photomask **410** includes a first photomask region **410***a* over the bottom electrode **312** of the MIM capacitor module **302***a*, and a second photomask **410***b* extending over the pair of metal heads **352***a*, **352***b* of the TFR module **304***a*. A lateral width W_{be} resist of the first photomask region **410***a* extends laterally across and beyond a lateral width W_{be} of the bottom electrode **312** in the x-direction, or in both the x-direction and y-direction. Similarly, a lateral width W_{tfr_resist} of the second photomask region **410***b* extends laterally across and beyond a lateral width W_{heads} extending across both metal heads **352***a*, **352***b* in the x-direction, or in both the x-direction and y-direction.

Next, as shown in FIG. **10**, an etch is performed to remove portions of the insulator layer **406** and the resistor film **404** uncovered by the patterned photomask **410**, leaving (a) a first resistor film region **404***a* and a first insulator layer region **406***a* below the first photomask region **410***a* and (b) a second resistor film region **404***b* and a second insulator layer region **406***b* below the second photomask region **410***b*. A lateral width of each of the first resistor film region **404***a* (W_{region_404a}) and the first insulator layer region **406***a* (W_{region_406a}) corresponds with the lateral width W_{be} resist of the first photomask region **410***a*, such that each extends laterally across and beyond the lateral width W_{be} of the bottom electrode in the x-direction, or in both the x-direction and y-direction. Similarly, a lateral width of each of the second resistor film region **404***b* (W_{region_404b}) and the second insulator layer region **406***b* (W_{region_406b}) corresponds with the lateral width W_{tfr_resist} of the second photomask region **410***b*, such that each extends laterally across and beyond the lateral width W_{heads} extending across both metal heads **352***a*, **352***b* in the x-direction, or in both the x-direction and y-direction.

Next an upper metal interconnect layer M_{x+1} is formed over the integrated circuit structure **300***a*. First, as shown in FIG. **11**, an upper metal layer **430**, e.g., comprising aluminum, copper, or other suitable metal, is deposited over the dielectric region **324** and the first and second insulator layer region **406***a*, **406***bb*.

Next, as shown in FIG. **12**, a photoresist layer is deposited and patterned to form a photomask **440** including (a) a first photomask element **440***a* for forming an upper interconnect contact **382**, as discussed below), (b) a second photomask element **440***b* (for forming the bottom electrode connection element **318**, as discussed below), (c) a third photomask element **440***c* (for forming the top electrode **316** and underlying insulator **314** and resistor film region **322**, as discussed below), and (d) a fourth photomask element **440***d* (for forming the metal resistor cap element **362** and underlying insulator cap element **360** and resistor element **354**, as discussed below).

As shown, a width W_{te_resist} of the third photomask element **440***c* may be larger than the width W_{be} of the bottom electrode **312** but smaller than the width of the first resistor film region **404***a* and first insulator layer region **406***a* (W_{region_404a}=W_{region_406a}), such that the subsequent metal etch to form the top electrode **316** (see FIG. **13**) self-aligns the lateral edges (in the x-direction, or both the x-direction and y-direction) of the top electrode **316** with the underlying insulator **314** and resistor film region **322** (see FIG. **13**), while also preventing any etching of the bottom electrode **312**.

Similarly, a width $W_{tfr\_cap\_resist}$ of the third photomask element **440***d* may be larger than the width W_{heads} extending across both metal heads **352***a*, **352***b* but smaller than the width of the second resistor film region **404***b* and second insulator layer region **406***b* (W_{region_404b}=W_{region_406b}), such that the subsequent metal etch to form the metal resistor cap element **362** (see FIG. **13**) self-aligns the lateral edges (in the x-direction, or both the x-direction and y-direction) of the metal resistor cap element **362** with the underlying resistor element **354** and insulator cap element **360** (see FIG. **13**), while also preventing any etching of the metal heads **352***a*, **352***b*.

FIG. **13** shows the results of the metal etch through the photomask **440** shown in FIG. **12**, and after removal of the photomask **440**, e.g., by a resist strip process. The etch may be selective to etch through structures not covered by the photomask **440** (including portions of the upper metal layer **430**, first and second insulator layer regions **406***a*, **406***b*, and portions of the first and second resistor film regions **404***a*, **404***b* not covered by the photomask **440**, as shown in FIG. **12**), but stop in the dielectric region **324**, e.g., at a depth of 500-1000 Å into the dielectric region **324**, for example. As shown, the etch forms (a) the upper interconnect element **382** in contact with the interconnect via **384**, (b) the bottom electrode connection element **318** in contact with the bottom electrode contact **320**, (c) the top electrode **316** and underlying insulator **314** and underlying resistor film region **322**, and (d) the metal resistor cap element **362** and underlying insulator cap element **360** and resistor element **354**.

As shown, as a result of the metal etch, the lateral edges of the top electrode **316** are aligned with corresponding lateral edges of the insulator **314** and the resistor film region **322**. A lateral width of the top electrode **316** (W_{te}), the insulator **314** (W_{ins}), and the resistor film region **322** ($W_{resistor\_film\_region}$) each extends laterally across and beyond the lateral width W_{be} of the bottom electrode in the x-direction, or in both the x-direction and y-direction.

Similarly, the lateral edges of the metal resistor cap element **362** are aligned with corresponding lateral edges of the underlying insulator cap element **360** and resistor element **354**. A lateral width of metal resistor cap element **362** (W_{metal_cap}), the insulator cap element **360** (W_{ins_cap}), and the resistor element **354** ($W_{tfr\_element}$) each extends laterally across and beyond the lateral width W_{heads} extending across both metal heads **352***a*, **352***b* in the x-direction, or in both the x-direction and y-direction.

Each of the resistor film region **322** and the resistor element **354** has the thickness ($T_{resistor}$) of the resistor film **404** from which they are formed, which thickness $T_{resistor}$ is uniform across the full lateral width (in the x-direction, or in both the x-direction and y-direction) of the resistor film region **322** and resistor element **354**, respectively. Similarly, each of the insulator **314** and the insulator cap element **360** has the thickness (T_{ins}) of the insulator layer **406** from which they are formed, which thickness T_{ins} is uniform across the full lateral width (in the x-direction, or in both the x-direction and y-direction) of the insulator **314** and insulator cap element **360**, respectively.

In some examples, FIG. **13** shows the completed of MIM capacitor module **302***a*, TFR module **304***a* and interconnect structure **306***a*. After forming the MIM capacitor module **302***a*, TFR module **304***a* and interconnect structure **306***a* the process may continue with construction of additional interconnect structures, e.g., by forming further metal interconnect layers and/or dielectric layers.

FIG. **14** is a flowchart showing an example method **1400** of forming an example integrated circuit structure including an example MIM capacitor module and an example TFR module, according to one example. At **1402**, a lower metal layer M_x is formed, including a bottom electrode base. In one example, the lower metal layer M_x comprise a lower interconnect metal layer (e.g., comprising aluminum). In another example, the lower metal layer M_x comprises a silicided polysilicon layer. At **1404**, a dielectric region (e.g., an oxide region) is deposited over the lower metal layer M_x, and planarized, for example by a CMP process.

Next, as indicated by the dashed line box in FIG. **14**, a damascene process may be performed at **1406-1412** to form a bottom electrode and a bottom electrode contact of the MIM capacitor module and a pair of metal heads of the TFR module.

First, at **1406**, the dielectric region is patterned and etched to form a tub opening and a pair of head openings. At **1408**, a bottom electrode is formed in the bottom electrode tub opening, the bottom electrode including (a) a cup-shaped bottom electrode component and (b) a bottom electrode fill component in an interior opening defined by the cup-shaped bottom electrode component.

At **1410**, which may be performed concurrently with **1408**, a metal head is formed in each of the pair of head tub openings, each metal head including (a) a cup-shaped head component; and (b) a head fill component in an interior opening defined by the cup-shaped head component.

At **1412**, a planarization process is performed to define a planarized support surface including (a) a planarized top surface of the cup-shaped bottom electrode component, (b) a planarized top surface of the bottom electrode fill component, and (c) planarized top surface areas of the dielectric region on opposite sides of the bottom electrode.

At **1414**, a resistor element connected across the pair of metal heads is formed from a resistor film formed on the planarized support surface.

At **1416**, an insulator is formed on the bottom electrode of the MIM capacitor module. The insulator may be formed over the planarized support surface, and may extend laterally across and beyond a full width of the bottom electrode, such that the insulator extends over the planarized top surface areas of the dielectric region on opposite sides of the bottom electrode, in the x-direction or in both the x-direction and y-direction. The insulator may have a uniform vertical thickness across a full lateral width of the insulator. In some examples, as explained above, the insulator is formed over the resistor film, such that a resistor film region (formed on the planarized support surface) is arranged between the insulator and the planarized support surface. Like the insulator, the resistor film region may have a uniform vertical thickness across a full lateral width of the resistor film region.

At **1418**, an upper metal layer is formed over the dielectric region, the upper metal layer including a top electrode over the insulator.

FIGS. **15**A-**15**B illustrate a flowchart showing an example method **1500** of forming an example integrated circuit structure including an example MIM capacitor module and a TFR module, according to one example. At **1502**, a lower metal layer M_x is formed, including a bottom electrode base and a pair of head bases. In one example, the lower metal layer M_x comprise a lower interconnect metal layer (e.g., comprising aluminum). In another example, the lower metal layer M_x comprises a silicided polysilicon layer. At **1504**, a dielectric region (e.g., an oxide region) is deposited over the lower metal layer M_x, and planarized, for example by a CMP process.

Next, as indicated by the dashed line box in FIG. **15**A, a damascene process may be performed at **1506-1512** to form a bottom electrode and a pair of metal heads. At **1506**, the dielectric region is patterned and etched to form (a) a tub opening and a pair of head openings. At **1508**, a conformal metal (e.g., tungsten) is deposited and extends into the bottom electrode opening and the pair of head openings. At **1510**, a fill metal (e.g., titanium nitride (TiN)) is deposited over the conformal metal and extends into the tub opening and the pair of head openings.

At **1512**, a planarization process is performed to remove portions of conformal metal and fill metal, wherein (a) remaining portions of conformal metal and fill metal in bottom electrode opening define a bottom electrode including a cup-shaped bottom electrode component and a bottom electrode fill component in an interior opening defined by the cup-shaped bottom electrode component and (b) remaining portions of conformal metal and fill metal in each head opening define a metal head including a cup-shaped head component and a head fill component in an interior opening defined by the cup-shaped head component. The planarization process may define a planarized support surface including (a) a planarized top surface of the cup-shaped bottom electrode component, (b) a planarized top surface of the bottom electrode fill component, (c) planarized top surface areas of the dielectric region on opposite sides of the bottom electrode, (d) a planarized top surface area of the cup-shaped head component of each metal head, (e) a planarized top surface area of the fill component of each metal head, and (f) planarized top surface areas of the dielectric region laterally between and adjacent the pair of metal heads.

At **1514**, a resistor element connected across the pair of metal heads is formed from a resistor film formed on the planarized support surface.

At **1516**, an insulator is formed over the bottom electrode. The insulator may be formed over the planarized support surface, and may extend laterally across and beyond a full width of the bottom electrode, such that the insulator extends over the planarized top surface areas of the dielectric region on opposite sides of the bottom electrode (in x-direction or in both x-direction and y-direction). In some examples, the insulator may be formed as a planar insulator having a uniform vertical thickness across a full lateral width of the insulator. In some examples, as explained above, the insulator is formed over the resistor film, such that a resistor film region (arranged on the planarized support surface) is arranged between the insulator and the planarized support surface. Like the insulator, the resistor film region may have a uniform vertical thickness across a full lateral width of the resistor film region.

At **1518**, an upper metal layer is formed over the dielectric region, the upper metal layer including a top electrode over the insulator.

The invention claimed is:

1. An integrated circuit structure, comprising:

(a) a metal-insulator-metal capacitor module, comprising:
  - a bottom electrode base formed in a lower metal layer;
  - a bottom electrode formed in a dielectric region between the lower metal layer and an upper metal layer, the bottom electrode comprising:
    - a cup-shaped bottom electrode component; and
    - a bottom electrode fill component formed in an interior opening defined by the cup-shaped bottom electrode component;
  - an insulator formed over the bottom electrode; and
  - a top electrode formed in the upper metal layer over the insulator; and

(b) a thin-film resistor module, comprising:
  - a pair of metal heads formed in the dielectric region between the lower metal layer and the upper metal layer, wherein each metal head comprises:

a cup-shaped head component; and

a head fill component formed in an interior opening defined by the cup-shaped head component; and

a resistor element connected across the pair of metal heads;

wherein the cup-shaped bottom electrode component of the bottom electrode and the cup-shaped head component of each metal head comprise respective portions of the same conformal metal layer comprising a conformal metal; and

wherein the bottom electrode fill component of the bottom electrode and the head fill component of each metal head comprise respective portions of the same fill metal layer comprising a fill metal different than the conformal metal; and

(c) an interconnect via extending vertically through the dielectric region between the lower metal layer and the upper metal layer, wherein the interconnect via is formed from a further respective portion of the conformal metal layer, and wherein the interconnect via is free of the fill metal.

2. The integrated circuit structure of claim **1**, wherein the thin-film resistor module includes a pair of head bases formed in the lower metal layer, wherein each metal head is conductively connected to a respective head base.

3. The integrated circuit structure of claim **1**, wherein the conformal metal comprises tungsten, and the fill metal comprises titanium nitride.

4. The integrated circuit structure of claim **1**, wherein the insulator has a uniform vertical thickness across a full lateral width of the insulator.

5. The integrated circuit structure of claim **1**, wherein the insulator is formed over a planarized support surface including a planarized top surface of the cup-shaped bottom electrode component and a planarized top surface of the bottom electrode fill component.

6. The integrated circuit structure of claim **5**, comprising a resistor film region formed on the planarized support surface, the resistor film region having a uniform vertical thickness across a full lateral width of the resistor film region;

wherein the insulator is formed on the resistor film region, the insulator having a uniform vertical thickness across a full lateral width of the insulator.

7. The integrated circuit structure of claim **1**, wherein:

the insulator is formed on a planarized support surface including (a) a planarized top surface of the cup-shaped bottom electrode component, (b) a planarized top surface of the bottom electrode fill component, and (c) planarized top surface areas of the dielectric region on opposite sides of the bottom electrode; and

the insulator extends laterally across and beyond a full width of the bottom electrode, such that the insulator extends over the planarized top surface areas of the dielectric region on opposite sides of the bottom electrode.

8. The integrated circuit structure of claim **1**, wherein:

the thin-film resistor module includes a resistor element insulator cap element formed on the resistor element; and

the resistor element insulator cap element is formed from a common insulator material as the insulator.

9. The integrated circuit structure of claim **1**, wherein:

the lower metal layer comprises a lower interconnect layer; and

the upper metal layer comprises an upper interconnect layer.

10. The integrated circuit structure of claim **1**, wherein:

the lower metal layer comprises a silicided polysilicon layer, wherein the bottom electrode base formed in the lower metal layer comprises a metal silicide region formed on a polysilicon region; and

the upper metal layer comprises a first metal interconnect layer.

11. The integrated circuit structure of claim **1**, wherein the metal-insulator-metal capacitor module comprises:

a bottom electrode connection element formed in the upper metal layer; and

a bottom electrode contact formed in the dielectric region between the lower metal layer and the upper metal layer;

wherein the bottom electrode contact provides a conductive connection between the bottom electrode connection element formed in the upper metal layer and the bottom electrode base formed in the lower metal layer.

12. The integrated circuit structure of claim **1**, further comprising an interconnect structure including:

a lower interconnect element formed in the lower metal layer; and

an upper interconnect contact formed in the upper metal layer and connected to the lower interconnect element by at least the interconnect via.

13. The integrated circuit structure of claim **12**, wherein the at least one interconnect via and the cup-shaped bottom electrode component are formed from a common conformal metal.

14. An integrated circuit structure, comprising:

(a) a metal-insulator-metal capacitor module, comprising:

a bottom electrode formed in a dielectric region, the bottom electrode comprising:

a cup-shaped bottom electrode component; and

a bottom electrode fill component formed in an interior opening defined by the cup-shaped bottom electrode component;

an insulator formed over the bottom electrode; and

a top electrode formed over the insulator; and

(b) a thin-film resistor module, comprising:

a pair of metal heads formed in the dielectric region, wherein each metal head comprises:

a cup-shaped head component; and

a head fill component formed in an interior opening defined by the cup-shaped head component; and

a resistor element connected across the pair of metal heads;

wherein the cup-shaped head component of each metal head and the cup-shaped bottom electrode component comprise respective portions of the same conformal metal layer comprising a conformal metal deposited in respective openings in the dielectric region; and

wherein the head fill component of each metal head and the bottom electrode fill component comprise respective portions of the same fill metal layer comprising a fill metal different than the conformal metal; and

(c) an interconnect via extending vertically through the dielectric region between the lower metal layer and the upper metal layer, wherein the interconnect via is formed from a further respective portion of the conformal metal layer, and wherein the interconnect via is free of the fill metal.

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